INCH-POUND

MIL-T-53066B
28 September 1994
SUPERSEDING
MIL-T-53066A
8 May 1990

MILITARY SPECIFICATION

TANK, FABRIC, COLLAPSIBLE, 5,000 BARREL, FUEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 5,000 barrel capacity, collapsible, self-supporting, fuel storage tank, complete with fittings, accessories and emergency repair items.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging.
L-P-378	-	Plastic, Sheet and Strip, Thin Gauge, Polyolefin.
RR-C-271	-	Chains and Attachments, Welded and Weldless.
VV-F-800	-	Fuel Oil, Diesel.
WW-V-35	-	Valve, Ball.
ZZ-H-561	-	Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-C-795	-	Cushioning Material, Packaging (Flexible Closed Cell Plastic Film for Long Shipping Cycle Applications).
PPP-C-1797	-	Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-T-60	-	Tape: Packaging, Waterproof.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN AMSTA RBE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5430

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MIL-C-104	-	Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	-	Preservation, Methods of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-H-370	-	Hose and Hose Assemblies, Nonmetallic: Rubber, Liquid Fuel.
MIL-T-5624	-	Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-T-22085	-	Tapes, Pressure-Sensitive, Adhesive, Preservation and Sealing.
MIL-T-27730	-	Tape, Antiseize, Polytetrafluoroethylene, with Dispenser.
MIL-F-46162	-	Fuel, Diesel, Referee Grade.
MIL-R-52255	-	Repair Kit and Repair Kit Components, for Collapsible Fabric Tanks, Drums, and Boats.
MIL-V-58039	-	Valves, Gate, Rising Stem, Double Acting, Aluminum.
MIL-T-83133	-	Turbine Fuels, Aviation, Kerosene Types, NATO F-34 (JP-8) and NATO F-35.

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods.
FED-STD-595	-	Colors Used in Government Procurement.

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-209	-	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MS9021	-	Packing, Preformed, "O" Ring.
MS27022	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Male, External Pipe Thread, Type III.
MS27023	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Male, Flanged, Type IV.
MS27024	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Internal Pipe Thread, Type V.
MS27026	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Female, External Pipe Thread, Type VII.
MS27027	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Female, Flanged, Type VIII.
MS27028	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Cap, Dust, Type IX.
MS27029	-	Coupling Half, Quick Disconnect, Cam-Locking Type, Plug, Dust, Type X.
MS27030	-	Gasket-Coupling Half, Quick Disconnect, Cam-Locking Type.
MS29513	-	Packing, Preformed, Hydrocarbon Fuel Resistant "O" Ring.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.14	-	Ferrous Pipe Plugs, Bushings, and Locknuts with Pipe Threads.
B16.15	-	Cast Bronze Thread ed Fitting Class 125 and 250.
B18.2.1	-	Square and Hex Bolts and Screws Inch Series.
B18.2.2	-	Square and Hex Nuts (Inch Series).
B18.21.1	-	Lock Washers.
B18.22.1	-	Plain Washers.
H35.1	-	Alloy and Temper Designation Systems for Aluminum.

(Application for copies should be addressed to the AMERICAN NATIONAL STANDARDS INSTITUTE, 11 W 42ND ST, NEW YORK, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 153	-	Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
B 26	-	Aluminum-Alloy Sand Castings.
B 85	-	Aluminum-Alloy Die Castings.
B 108	-	Aluminum-Alloy Permanent Mold Castings.
D 381	-	Existent Gum in Fuels by Jet Evaporation.
D 412	-	Rubber Properties in Tension.
D 413	-	Rubber Property - Adhesion to Flexible Substrate.
D 429	-	Rubber Property - Adhesion to Rigid Substrates.
D 471	-	Rubber Property - Effect of Liquids.
D 750	-	Rubber Deterioration in Carbon-Arc or Weathering Apparatus.
D 751	-	Coated Fabrics.
D 975	-	Diesel Fuel Oils.
D 1149	-	Rubber Deterioration - Surface Ozone Cracking in a Chamber.
D 1655	-	Aviation Turbine Fuels.
D 1729	-	Visual Evaluation of Color Differences of Opaque Materials.
D 2000	-	Rubber Products in Automotive Applications.
D 2565	-	Operating Xenon Arc-Type (Water-Cooled) Light-Exposure Apparatus With and Without Water for Exposure of Plastics.

(Application for copies of ASTM documents should be addressed to: AMERICAN SOCIETY FOR TESTING & MATERIALS, 1916 RACE STREET, PHILADELPHIA PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or military standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. A complete unit shall consist of an elastomeric coated fabric tank with attached handles, fittings, accessories, and repair items (see 6.2 for ordering options).

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2.1 First article sample tests. In addition to the first article tank, the contractor shall make samples of coated fabric, seams, and fittings for testing as specified herein. The samples shall be fabricated during construction of the first article tank. Samples of cloth, coating compounds, coating adhesion samples and coated fabric to aluminum strip samples shall also be furnished for testing. The coating compounds, coated fabric, coating adhesion and coated fabric to aluminum strip samples shall be processed

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identically (time, temperature, pressure, etc.) to the materials in the first article model. The contractor shall also provide a small model tank as specified in 4.5.2.4, that is manufactured identically (same materials, production personnel, production procedures, equipment, etc.) to the first article tank.

3.2.1.1 Samples for submission. The contractor shall furnish with the first article test report samples of coated fabric and samples of seams. The coated fabric samples shall be a minimum of one square yard. The seam sample shall be a minimum of 36 inches of seam with a minimum of 6 inches of coated fabric on either side of the seam. Samples shall be fabricated using production personnel and techniques. The samples shall be sent to: US ARMY BELVOIR RDE CTR, ATTN AMSTA RBWH, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818. Samples shall be marked with the appropriate contract number.

3.2.2 Process data. Within the time frame specified (see 6.2), the contractor shall make process data available to the Government (in accordance with 6.3), at the manufacturing site.

3.3 Materials. Materials shall be as specified in text and figures herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Tanks may not be made with materials that fail to meet the requirements of tables I through IV; however, conformance with these requirements shall not be construed as justification for failure to meet other performance requirements of this specification.

3.3.1 Material deterioration prevention and control. The tanks shall be fabricated from compatible materials, inherently deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the tanks may be exposed.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The metal components, pieces and parts incorporated in the tank may be newly fabricated from recovered materials to the maximum extent practicable, provided the tank produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the tanks.

3.3.3 Cloth. When tested as specified in 4.5.2.23, the cloth of the coated fabric shall have a retained breaking strength that is not less than 50 percent of its initial breaking strength, warp and fill.

3.3.4 Coating compounds. Elastomeric compound applied to the cloth and seam edge cover strips (if used) shall conform to the requirements of table I.

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TABLE I. Characteristics of coating compounds. 1/

Test Property	Requirements	Test paragraph and ASTM test methods
Original properties Tensile strength, initial Ultimate elongation	 1500 psi (min) 300% (min)	 D 412 D 412
Properties after fuel immersion in test fluid 2/ at 160 °F for 14 days Tensile strength retained Elongation retention Volume swell	 80% (min) 80% (min) 25% (max)	D 471 (para 14.1, 14.2, & 10.1)
Properties after fuel extraction, dried, and then immersed in distilled water at 160 °F for the following durations: 3/ 14 days Tensile strength retention Elongation retention Volume swell 42 days Tensile strength retention Elongation retention Volume swell	 75% (min) 80% (min) 10% (max) 70% (min) 75% (min) 10% (max)	D 471 (para 14.1, 14.2, & 10.1) & 4.5.2.19
Resistance to light after 1500 hours accelerated weathering at 10% elongation 4/ Tensile strength retention	 80% (min)	D 750 6/ or D 2565 7/
Fuel contamination: 5/ Existent gum, unwashed Heptane washed gum	 20 mg/100mL (max) 5 mg/100mL (max)	 4.5.2.9 4.5.2.9
Ozone resistance	No cracks under 7x lens	D 1149 8/

1/ Properties of coating compounds/slabs shall be identical in composition to the coating compounds used on the coated fabric.

2/ Immersion test fluid shall be diesel fuel conforming to MIL-F-46162.

3/ JP-5/JP-8 ST conforming to MIL-T-5624 will be used as extraction media. If JP-5/JP-8 ST is unavailable at a reasonable cost and within time constraints, the manufacturer can use JP-8 with enough BTX added to ensure aromatics equal or exceed 23 percent.

4/ Applicable to all exterior coating compounds. That is, all coating compounds between the cloth and outside of the tank.

5/ Applicable to all interior coating compounds and seam covering materials. That is, coating compounds between the nylon cloth (including any coatings or seam covering tapes) and the inside of the tank.

6/ Alternate Corex D filters in place.

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7/ ASTM Method D 2565, Xenon Light, Procedure A, inner and outer borosilicate filters; deionized water (20 ±3 °C); cycle: 690 minutes light exposure, 30 minutes light and gray; black panel temperature 63 ±3 °C; relative humidity (45 ±5%).

8/ Test method A specimen shall be conditioned for 14 days at a temperature of 104 ±3.6 °F (40 ±2 °C) having a partial pressure of ozone of 50 milipascals.

3.3.5 Coated fabric. The processed coated fabric shall be free from blisters, pinholes, or signs of coating delamination. The processed coated fabric shall conform to the requirements of table II.

TABLE II. Characteristics of coated fabric.

Test property	Requirements	Test paragraph, test method of FED-STD-191 or ASTM test method
Weight (oz/sq yd)	40 min, 62 max	5041
Diffusion rate 1/ fl oz/sq ft/24 hr, max.	0.12	4.5.2.10
Tear strength, warp & fill lb., minimum	50	5134
Breaking strength, warp & fill, lb/in, min	600	5102
Puncture resistance lbs., minimum	225	4.5.2.11/5120
Weathering resistance, 1500 hrs at 5% elong, warp & fill breaking strength retention, % min. 5/	80	5804 3/ or D 2565 4/ and 5102
Low temperature crease resistance:		
Appearance after unfolding	No cracking, peeling, or delamination under 7x lens	4.5.2.12
Diffusion rate fl oz/sq ft/24 hrs, max 1/	0.12	4.5.2.10
Blocking	Specimens to separate within 5 seconds	4.5.2.13
Coating adhesion initial, lb/in, min	30	4.5.2.14 and 4.5.2.14.1
After fuel immersion 2/ for 14 days at 160 °F lb/in, min	20	D 471, 4.5.2.14 and 4.5.2.14.1
After fuel extraction 1/, dried, and immersed in distilled water at 160 °F for:		D 471, 4.5.2.14, 4.5.2.14.1, and 4.5.2.19
14 days, lb/in, min	20	
42 days, lb/in, min	15	

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1/ JP-5/JP-8 ST conforming to MIL-T-5624 will be used as extraction media. If JP-5/JP-8 ST is unavailable at a reasonable cost and within time constraints, the manufacturer can use JP-8 with enough BTX added to ensure aromatics equal or exceed 23 percent.

2/ Immersion test fluid shall be diesel fuel conforming to MIL-F-46162.

3/ Alternate Corex D filters in place. Coated fabric specimens shall have exterior coating (outside of tank) facing the carbon arc.

4/ ASTM D 2565, Xenon Light, Procedure A, inner and outer borosilicate filters; deionized water (20 ±3 °C); 690 minutes light exposure, 30 minutes light and gray; black panel temperature 63 ±3 °C; relative humidity (45 ±5%). Coated fabric specimens shall have exterior coating (outside of tank) facing the light.

5/ This requirement may be met by certification from the material manufacturer that the material provided has been subjected to, and passed the specified weather resistance test, and providing those test results for inclusion in the first article test report.

3.4 Tank construction. The tank shall be fabricated from coated fabric as specified herein. The configuration and dimensions of the tank and the location of the fittings and handles shall be as shown in figure 1. Seams shall not coincide with tank fittings. The longitudinal seams of the tank top shall not coincide with the seams of the tank bottom at each end closure. Coated fabric panels may be spliced together; however, all splices shall be located on the bottom of the tank. All splices shall conform to requirements for seams as specified in 3.4.1. Splices in adjacent panels shall not coincide. On both the interior and exterior of the tank, all coated fabric, edges of seams, fabric flanges of fittings, and splices shall be covered to a thickness of not less than 10 mils with coating compound. All tank fittings shall be located a minimum of 8 inches away from any seam or splice.

3.4.1 Seams. All tank seams including the end closures, fabric flanges of fittings, fabric panel splices and the bonded areas of chafing and handle patches shall conform to the requirements of table III. Seams shall be designed and manufactured such that fluids cannot enter them during tank life.

TABLE III. Characteristics of seams.

Test Property	Requirements	Test Paragraph or ASTM Test Method
Breaking strength:		4.5.2.15 and D 751 method B
Initial, lb/in, min 1/	600	
After immersion in fuel 2/ at 160 °F for:		4.5.2.15, D 751 method B, and D 471 (para. 15.2)
14 days, lb/in, min 1/	450	
After fuel extraction 3/, dried, and immersed in distilled water at 160 °F for:		4.5.2.15, 4.5.2:19, D 751 method B, and D 471 (para 15.2)
14 days, lb/in, min 1/	450	
42 days, lb/in, min 1/	400	
Dead load shear resistance under 50 lb/in stress at 180 °F for 8 hours	0.125 in. slippage (max)	4.5.2.16

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TABLE III. Characteristics of seams. - (continued)

Test Property	Requirements	Test Paragraph or ASTM Test Method
Seam peel adhesion		4.5.2.15 and D 413 machine method
Initial, lb/in, min	35	
After immersion in fuel 2/ at 160 °F for:		4.5.2.15, D 413 machine method, and D 471 (para 15.2)
14 days, lb/in, min	25	
After fuel extraction 3/, dried, and immersed in distilled water at 160 °F for:		4.5.2.15, 4.5.2.19, D 413 machine method, and D 471 (para 5.2)
14 days, lb/in, min	25	
42 days, lb/in, min	20	

1/ All specimens must break in the coated fabric. Failure of any specimen in a seam area at any value shall constitute failure of this test.

2/ Immersion test fluid shall be diesel fuel conforming to MIL-F-46162.

3/ JP-5/JP-8 ST conforming to MIL-T-5624 will be used as extraction media. If JP-5/JP-8 ST is unavailable at a reasonable cost and within time constraints, the manufacturer can use JP-8 with enough BTX added to ensure aromatics equal or exceed 23 percent.

3.4.1.1 Lap joints and butt joints. Lap joints or butt joints shall be used to fabricate seams between adjacent panels and splices. Lap joints shall have a minimum overlap length of 3 inches. Lap joints shall have a gum strip or adhesive barrier centered over the inner and outer exposed coated fabric edge to prevent wicking through the fabric. Butt joints shall have a 3-inch wide (minimum) patch centered over the butt joint on both sides of the coated fabric.

3.4.2 Handles. Handles shall be bonded to the tank as shown in figure 1. The handles shall be U-shaped straps fabricated from nylon webbing or cloth and fastened to a coated fabric patch similar to the tank body material. The two ends of each strap shall be attached to each patch at points 12 inches ±1 inch apart. The length of the strap between the two points of attachment shall be 16 inches ±1 inch. The patch and strap assembly shall be bonded to the bottom of the tank just below the peripheral fold line. The bonds between each handle patch assembly and the tank fabric shall be capable of withstanding perpendicular loads of 1,500 pounds without damage to the tank. They shall be approximately the same color as the tank.

3.4.3 Chafing patches. The interior and exterior of the tank, opposite the location of each fitting shall be provided with bonded coated fabric chafing patches as shown in figure 1. The chafing patches shall be made of the same coated fabric used to fabricate the tank.

3.4.4 Fittings.

3.4.4.1 Fitting type. The access door fitting and the vent and drain flange attachment shall be bonded to the coated fabric flange and shall be as specified herein and shall conform to the requirements of table IV. Fittings shall be compression type and shall be as shown in figures 3 and 6. The oval closure plate shall be as shown in figure 9.

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TABLE IV. Characteristics of bonded fittings.

Test Property	Requirements	Test Paragraph or ASTM Test Method
Breaking strength - (Aluminum to coated fabric)		4.5.2.17 and 4.5.2.17.1
Initial, lb/in, min	600	
After immersion in fuel 1/ at 160 °F for:		4.5.2.17 and 4.5.2.17.2
14 days, lb/in, min	450	
After fuel extraction 2/, dried, and immersed in distilled water at 160 °F for:		4.5.2.17, 4.5.2.17.2, and 4.5.2.19
14 days, lb/in, min	450	
42 days, lb/in, min	400	
Dead load shear resistance under 60 lb/in stress at 180 °F for 8 hours	0.125 in slippage (max)	4.5.2.17.3
Peel adhesion (Aluminum strip to coated fabric)		D 429 method B, 4.5.2.18, and 4.5.2.18.1
Initial, lb/in, min	30	
After immersion in fuel 1/ at 160 °F for:		D 429 method B, 4.5.2.18, and 4.5.2.18.1
14 days, lb/in, min	20	
After fuel extraction 2/, dried, and immersed in distilled water at 160 °F for:		D 429, method B, 4.5.2.18, 4.5.2.18.1 and 4.5.2.19
14 days, lb/in, min	20	
42 days, lb/in, min	15	

1/ Immersion test fluid shall be diesel fuel conforming to MIL-F-46162.

2/ JP-5/JP-8 ST conforming to MIL-T-5624 will be used as extraction media. If JP-5/JP-8 ST is unavailable at a reasonable cost and within time constraints, the manufacturer can use JP-8 with enough BTX added to ensure aromatics equal or exceed 23 percent.

3.4.4.2 Fitting assemblies. Unless otherwise specified (see 6.2), each tank shall be furnished with the following fitting assemblies located as shown in figure 1. The filler/discharge shall be assembled as shown in figure 2. The drain fitting shall be assembled as shown in figure 7. The vent fitting shall be assembled as shown in figure 5. The pressure relief cap of the vent assembly shall be open when subjected to an internal pressure of 3 inches of water. The recommended torque for all fitting bolts shall be either stamped onto the fitting flange plates or stenciled on the tank fabric adjacent to the respective fittings. Keeper chains shall be attached to all dust covers (caps and plugs) and their adjacent fittings such that the dust cover may be removed without detaching the keeper chain at either of its ends. The keeper chains shall be in accordance with RR-C-271, type II, class 3, bronze, nonreflective. All 1/4-inch bolts shall be torqued to a minimum of 5 ft lb and all 3/8-inch bolts shall be torqued to a minimum of 15 ft lb.

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3.4.4.3 Aluminum alloys of tank fittings. Alloy and temper designations of wrought aluminum shall be in accordance with ANSI H35.1. Cast aluminum alloy shall conform to ASTM B 85, UNS A14130, UNS A13600, or UNS A05180 for die castings; ASTM B 108, UNS A03560-T6 for permanent mold castings, or ASTM B 26, UNS A13560-T6 or UNS A07120-T5 for sand castings.

3.4.4.4 Protective coatings. The contractor shall provide certification that the minimum coating thickness for castings is at least 0.004 inches and at least 0.0007 inches for wrought aluminum alloy fittings.

3.4.4.5 Thread sealant. Antiseize tape conforming to MIL-T-27730A, size II, shall be applied to mating pipe threads prior to assembly (figures 5, 7, 20 & 21). A thread sealant shall be used on those threaded fasteners when those threaded fasteners are subject to leakage.

3.5 Tank performance. The tanks and components shall withstand folded storage at ambient temperatures from +160 to -30 °F without damage or leakage when subsequently filled. The tank and components shall be suitable for operational use at ambient temperatures from +125 to -25 °F. The tank shall be suitable for use in continuous contact with rainwater, ground water, or water associated with fuels stored in the tank. There shall be no evidence of leakage or seepage when the tank is filled to capacity of 5,000 barrels of fuel. The tank shall have a minimum over capacity of 500 barrels of fuel without rupture or evidence of weakened areas and without leakage or seepage of fuel. The tank shall be capable of withstanding an internal air pressure of 0.50 pound per square inch (psi) without evidence of leakage.

3.6 Accessories. Unless otherwise specified (see 6.2), each tank shall be provided with the following accessories:

ITEM	CONFORMANCE	QUANTITY
Hose Assembly (Drain)	Figure 20	2
Ball Valve Assembly (Drain)	Figure 21	2
Hose Assembly (Filler/Discharge)	Figure 22	4
Gate Valve Assembly (Filler/Discharge)	Figure 23	1
Lifting Sling	Each lifting sling shall be a minimum of 3 inches in width by 13 feet in length, of polyester, with minimum tensile strength of 5000 lbs, and be equipped with a delta ring on each end (see 5.2.5).	3
Deployment Strap	Each deployment strap shall be a minimum of 2 inches in width by 8 feet in length, of polyester or nylon, and have a minimum tensile strength of 500 lbs (see 5.2.5).	8

The drain hoses shall be assembled with the dust covers attached as shown in figure 20. The drain valves shall be assembled with the dust covers attached as shown in figure 21. The filler/discharge hoses shall be assembled with the dust covers attached as shown in figure 22. The filler/discharge valve shall be assembled with the dust covers as shown in figure 23. All dust covers shall be attached to the assembly with keeper chains that meet the requirements of 3.4.4.2.

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3.7 Repair items. The following emergency repair items shall be furnished with each tank:

Item	Quantity
Repair kit and repair kit components conforming to MIL-R-52255, type II	1 ea
O-Ring MS9021-383	2 ea
O-Ring MS29513-250	2 ea
Gasket, quick-disconnect coupling MS27030-6	8 ea
Gasket, quick-disconnect coupling MS27030-10	14 ea
Gasket, 6-inch flange, figure 10	6 ea
Tape, Antiseize, MIL-T-27730, size II	1 ea
One square yard of coated fabric conforming to tables I and II	1 ea

3.8 Labels. The letters for the various labels required for this tank shall be a minimum of 1-inch high. The letters width shall be 0.6 of the height except for "M" and "W", which shall be 0.8 of the height, and "I", which shall be one stroke wide. The numerals shall be the same height as the letters. The numeral width shall be 0.6 of the height except for the "4", which shall be 0.7 of the height and the "1" which shall be one stroke wide. The minimum space between characters shall be one stroke width. The minimum space between words shall be the width of one character. The minimum space between lines shall be one-half character height.

3.8.1 Tank caution label. The tank shall have a caution label bonded onto the body of the tank near each inlet as shown below. The word "CAUTION" shall be in 1.35-inch high letters. Bonded label shall be made of coating compound or coated fabric as specified in 3.3.5:

CAUTION

ATTACH DRAIN HOSE AND VALVE BEFORE FILLING TANK

DO NOT OVERFILL
TANK HOLDS 5,000 BARRELS (210,000 GALLONS)
MAXIMUM TANK HEIGHT IS 6 FT - 8 IN

NOT FOR LONG TERM GASOLINE STORAGE

3.8.2 Drain location labels. The chafing patch above each drain assembly shall be stenciled with the following message "DRAIN FITTING IS UNDER THIS LABEL." "CONNECT DRAIN HOSE BEFORE FILLING TANK." Minimum letter size is 1 inch.

3.8.3 Filler/discharge labels. "FILL/DISCHARGE" shall be stenciled above each filler discharge assembly in 1-inch (minimum) letters.

3.8.4 Identification label. The tank shall be identified by means of an identification label located as shown in figure 1. The label shall be made of coating compound or coated fabric bonded to the tank. The following information shall be molded, either recessed or in relief using 1/2-inch (minimum) letters, on the tank identification label:

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TANK, FABRIC, COLLAPSIBLE:
5,000 BARREL FUEL
NSN: (Specify)
SERIAL NO.: (Specify)
MFR: (Specify manufacturer's name and location of plant)
MFG DATE: (Specify month and year)
WEIGHT EMPTY: (Specify approximate no. of pounds)
CONTRACT OR ORDER NO.: (Specify)

3.9 Workmanship. The fabric-reinforced flange-type fittings shall contain no gum voids, cracks or tears that could adversely affect the strength of the assembly. All metal parts shall be clean and free of sand, dirt, and scale, flux, burrs, sharp edges, corrosion, and shall not be broken or malformed. Metal surfaces shall be smooth with edges rounded or beveled. The inside and outside of the tank shall be clean and free of foreign materials (excluding talc). Any necessary rework shall meet all applicable requirements of this specification. The cemented surfaces of all spliced areas, fitting flanges, and patch-type repairs shall effect a bond that will result in strength of the cemented area not less than the strength of adjacent tank fabric. Fabric components shall be free of holes, cuts or tears, thin, or weak areas, caused by abrasion or delamination, exposed fabric, blisters, tunnels, unadhered pockets, picks, loose edges, or any delamination of coating.

3.10 Repair and rework of collapsible fabric tanks. Any repair or rework of any tank shall be accomplished before inspection (quality conformance inspection or inspection of packaging, as applicable). Repair of seams shall be limited such that total repairs do not exceed ten percent of total seam length in the tank. Defects subject to repair (other than those on seams) shall be limited to six inches in diameter, six inches in length, and six inches in width as applicable. The 6-inch criteria shall be the maximum dimension of the affected condition.

3.11 Color When tested as specified in 4.5.2.20, the color of the tank exterior (the coating of the coated fabric, exposed fittings, handles, seams, and chafing patches) and hose assemblies shall approximate color 33446 of FED-STD-595.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order and unless disapproved by the Government, the contractor's own or any other facilities suitable for the performance of the inspection requirements specified herein, may be used. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection. The first article inspection shall be performed on one complete tank assembly (see 3.2 and 6.5).

4.3.1 Examination. The tank shall be examined as specified in table V and 4.6.2.2. Any nonconformance revealed by this examination shall be cause for rejection.

4.3.2 Tests. The tank and samples shall be subjected to test marked "X" in column 1 of table VI. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for quality conformance tests. The frequency of sampling shall be once for each 25 tanks produced. The 25 tanks shall be identified as a lot corresponding to the representative samples. Tests shall be performed on representative samples. The samples shall be manufactured concurrently with the corresponding lot using the same materials, processes, procedures, equipment, and production personnel.

4.4.2 Examination. Each complete tank assembly shall be examined as specified in table V. Any nonconformance revealed by the examination shall be cause for rejection of the tank.

4.4.3 Sample tests. Samples and the tank selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 2 of table VI. Failure of any test shall be cause for rejection of the represented lot of tanks.

4.5 Inspection procedure.

4.5.1 Examination schedule.

TABLE V. Examination schedule.

Examination number	Examination description	Requirement paragraph
101.	Samples for submission not as specified (applies to first article examination only).	3.2.1.1
102.	Process data not as specified.	3.2.2
103.	Materials not as specified	3.3
104.	Materials not resistant to corrosion and deterioration, or treated to be made resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
105.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the tanks.	3.3.2
106.	Blisters or pinholes in coated fabric.	3.3.5
107.	Location of splices not as specified.	3.4
108.	Dimensions not as specified.	3.4
109.	Splices and seams not coincident or seams coincident with tank fittings.	3.4
110.	Edges or coated fabric not covered as specified.	3.4
111.	Lap joints or butt joints not as specified.	3.4.1.1
112.	Handles not as specified.	3.4.2
113.	Chafing patches not as specified.	3.4.3
114.	Fittings not as specified.	3.4.4 thru 3.4.4.5
115.	Accessories not as specified.	3.6
116.	Repair items missing or not as specified.	3.7

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TABLE V. Examination schedule. - (continued)

Examination number	Examination description	Requirement paragraph
117.	Labels not as specified.	3.8 thru 3.8.4
118.	Workmanship not as specified.	3.9
119.	Extraneous material inside tank.	3.9
120.	Fittings, chafing patches, or handles not located as specified.	Figure 1
121.	O-rings not lubricated as specified.	Figs. 2, 5, 7, 20, 21, 22 & 23
122.	Repair and rework not as specified.	3.10

4.5.2 Test.

4.5.2.1 Test schedule.

TABLE VI. Test schedule.

First article	Quality Conform	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
		Tank and accessories		
X	-	Vent assembly	4.5.2.2	3.4.4.1
X	X	Air-leakage	4.5.2.3	3.5
X	-	Environmental testing	4.5.2.4	3.5
X	-	Storage test	4.5.2.5	3.5
X	-	Overload	4.5.2.6	3.5
X	-	Internal inspection	4.5.2.7	3.5
X	-	Handle pull test	4.5.2.8	3.4.2
		Cloth		
X	-	Weathering resistance	4.5.2.23	3.3.3

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TABLE VI. Test schedule. - (continued)

First article	Quality Conform	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
		Coating compounds		
X	-	Tensile strength	4.5.2.21	3.3.4, table I
X	-	Ultimate elongation	4.5.2.21	3.3.4, table I
		Fuel immersion:	4.5.2.21	3.3.4, table I
		After 14-days		
X	-	• Tensile strength		
X	-	• Elongation		
X	-	• Volume swell		
		Fuel extraction:	4.5.2.21	3.3.4, table I
		After 14 days		
X	-	· Tensile strength		
X	-	· Elongation		
X	-	· Volume swell		
		After 42 days		
X	-	· Tensile strength		
X	-	· Elongation retention		
X	-	· Volume swell		
X	-	Resistance to light	4.5.2.21	3.3.4, table I
X	-	Existent gum	4.5.2.9	3.3.4, table I
X	-	Ozone resistance	4.5.2.21	3.3.4, table I
		Coated fabric		
X	X	Weight	4.5.2.22	3.3.5, table II
X	X	Diffusion rate	4.5.2.10	3.3.5, table II
X	X	Tearing strength	4.5.2.22	3.3.5, table II
X	X	Breaking strength	4.5.2.22	3.3.5, table II
X	-	Weathering resistance	4.5.2.22	3.3.5, table II
X	X	Puncture resistance	4.5.2.11	3.3.5, table II
X	X	Low temperature crease resistance	4.5.2.12	3.3.5, table II
X	X	Blocking	4.5.2.13	3.3.5, table II
		Coating adhesion	4.5.2.14	3.3.5, table II
X	X	· Initial		
		Fuel immersion after	4.5.2.14.1	3.3.5, table II
X	-	· 14 days		
		Fuel extraction after	4.5.2.14.1	3.3.5, table II
X	-	· 14 days		
X	-	· 42 days		

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TABLE VI. Test schedule. - (continued)

First article	Quality Conform	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5
		Seams		
		Breaking strength	4.5.2.15	3.4.1, table III
X	X	· Initial		
		Fuel immersion after	4.5.2.15	3.4.1, table III
X	-	· 14-days		
		Fuel extraction after	4.5.2.15	3.4.1, table III
X	-	· 14 days		
X	-	· 42 days		
X	X	Dead load shear resistance	4.5.2.16	3.4.1, table III
		Peel adhesion	4.5.2.15	3.4.1, table III
		· Initial		
X	X	Fuel immersion after	4.5.2.15	3.4.1, table III
		· 14-days		
X	-	Fuel extraction after	4.5.2.15	3.4.1, table III
		· 14 days		
X	-	· 42 days		
X	-			
		Bonded fittings		
X	X	Bond strength	4.5.2.17.1	3.4.4.1, table IV
		· Initial		
X	-	Fuel immersion after	4.5.2.17.2	3.4.4.1, table IV
		· 14 days		
X	-	Fuel extraction after	4.5.2.17.2	3.4.4.1, table IV
X	-	· 14 days		
		· 42 days		
X	X	Dead load shear resistance	4.5.2.17.3	3.4.4.1, table IV
X	X	Peel adhesion	4.5.2.18	3.4.4.1, table IV
		· Initial		
X	-	Fuel immersion after	4.5.2.18	3.4.4.1, table IV
		· 14 days		
X	-	Fuel extraction	4.5.2.18	3.4.4.1, table IV
X	-	· 14 days		
		· 42 days		
		Color		
X	-	Color	4.5.2.20	3.11

4.5.2.2 Vent assembly. Subject the vent assembly to an internal pressure of 3 inches of water. Inability of the pressure relief cap to open at this pressure shall constitute failure of this test.

4.5.2.3 Air leakage. Pressurize each tank to a 0.50 psi, ±5 percent, internal air pressure and allow to stand for 30 minutes. Then adjust the internal air pressure to 0.50 psi, ±5 percent. Then using a soap and water solution examine all the tank fabric, fittings, and external seams for leakage. Any evidence of air leakage shall constitute failure of this test.

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4.5.2.4 Environmental testing. Environmental tests at extreme high and low temperature limits will be conducted on a small model tank. A water leakage test will be performed after the low and high temperature tests have been successfully completed. The small model tank will be 250 gallons in capacity, 80-inch square, with only a vent fitting provided in the center of the tank terminating with a quick-connect dust cap (items 1, 2, 3, 4, 5, 7 and 11 of figure 5). The coated fabric and seam fabrication techniques shall be identical to the 5,000 barrel tank. The small model tank will be folded and rolled in a manner to duplicate types of folds on the packaged 5,000 barrel tank (see 5.2.5); then, it will be placed in an appropriate environmental chamber under weights which will duplicate loads which will be imposed on the bottom of the 5,000 barrel tank in its shipping container. Load calculations will be furnished as part of the test data.

4.5.2.4.1 Low temperature. The small model tank, folded and under weights as specified in 4.5.2.4, shall be placed in a low temperature environment of -30 ±2 °F for a period of 24 hours. The temperature shall then be increased to -25 ±2 °F for an additional 22 hours. At the end of this period, while still at -25 °F, the tank shall be completely unfolded and examined. Any evidence of flaking, cracking, or separation of a seam or coated fabric observed during the examination shall constitute failure of this test.

4.5.2.4.2 High temperature. After successfully completing the low temperature test, the small model tank, folded and under weights as specified in 4.5.2.4, shall be placed in a high temperature environment of 160 ±2 °F for a period of 24 hours. The temperature shall then be decreased to 125 ±2 °F for an additional 12 hours. At the end of this period, while still at 125 °F, the tank shall be completely unfolded. The tank shall be removed from the high temperature chamber and filled with 250 gallons of water and allowed to stand undisturbed for 6 hours. At the end of this period, while still filled with water the tank shall be examined. Any evidence of leakage, flaking, cracking, or separation of a seam or coated fabric observed during the examination shall constitute failure of this test.

4.5.2.5 Storage test. Fill the tank with 5,000 barrels of one of the following fuels: military fuel (JP-5 per MIL-T-5624 or JP-8 per MIL-T-83133), commercial jet fuel (per ASTM D 1655), or diesel fuel (per VV-F-800 or ASTM D 975) and allow to stand for 30 days, +2 -0 days. At the end of 30 days, +2 -0 days, examine the tank for seepage and leakage. Any evidence of leakage or seepage shall constitute failure of this test.

4.5.2.6 Tank overload. At the conclusion of the storage test, an additional 500 barrels of the same fuel selected for use in 4.5.2.5 shall be added to the tank and allowed to stand for 4 hours. Any evidence of rupture, weakened areas, leakage or seepage shall constitute failure of this test.

4.5.2.7 Internal inspection. The tank shall be examined internally after the overload test. Any evidence of weakened areas, coating or barrier delamination shall constitute failure of the overload test.

4.5.2.8 Tank handle pull resistance. The test sample shall consist of the handle, coated fabric patch, and a minimum of 1 foot of tank body fabric extending in all directions from the handle patch. The body fabric shall be tightly drawn and clamped between two flat oval rings or a base plate in an oval ring so that the handle of the patch is centrally located in the rings. The shape of the oval rings shall be such that all edges of the coated fabric patch are a minimum of 1 inch from the ring clamp. The rigidity, strength and construction of the clamp shall be such that the tank body material shall not slip more than 1/2 inch at any point during the test. With the sample held securely, a tension shall be applied through a bar or pipe 1 inch in diameter, inserted in the loop of the handle. The tension shall be applied slowly and smoothly in a direction perpendicular to the plane of the handle patch until the specified load of 1,500 pounds is reached. The 1,500-pound load shall be maintained for 1 minute. Any damage, permanent distortion, or separation of the handle patch, or tank material shall constitute failure of this test.

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4.5.2.9 Existent gum. Cut a 5-gram specimen of each tank interior coating compound into approximately 0.0625-inch squares and place in a flask containing 250 ml of JP-5/JP-8 ST fuel conforming to note 3/ of table I and a flask containing MIL-F-46162 diesel fuel, and allow to stand for 48 hours at 73 ±5 °F. Decant and filter the contaminated fluid through Whatman 41H filter paper or equal. Determine the unwashed gum content of the filtrate in accordance with ASTM D 381, procedures 10.1 through 10.7 using the air jet or steam jet vaporizing medium (whichever is appropriate for the fuel) and an evaporation time of 45 minutes. Using the same samples after completing the unwashed gum content test above, determine the existent gum in accordance with ASTM D 381, procedures 10.8 through 10.12. A minimum of three specimens shall be used to report the average gum content. Nonconformance to 3.3.4 and table I shall constitute failure of this test.

4.5.2.10 Diffusion. The test apparatus shall consist of a diffusion cup and ring, using figure 24 as a guide. Other cup designs are acceptable as long as the inside diameter of the cup and ring is maintained at 2 inches ±0.016 inch and the cup is 1-inch deep (minimum). Cut a circular test disk of coated fabric to conform to the outside diameter of the cup flange. Punch holes in the disk to correspond to the flange bolt dimensions as needed. The cup shall be filled with approximately 40 ml of fuel conforming note 1/ of table II. A suitable solution (see 6.4), shall be used to seal the exposed fabric around the outer edge of the test disk and also for sealing the test disk to the diffusion cup flange. The test disk shall be placed over the cup with the tank "interior" side towards the fuel. The bolts shall be tightened securely. Place the diffusion cup in a suitable rack in a constant temperature of 73.3 ±2 °F, and a relative humidity of 65 ±2 percent. Allow 1 hour for the assembly to reach equilibrium, then weigh the cup to the nearest 0.005 gram and place in the rack face upward. Keep the cup at the above constant temperature and humidity for 24 hours. Then weigh and check for vapor loss. Retorque the bolts if necessary. Invert the cup (test disk down) in a rack that permits free access of air to the test disk. Weigh the cup daily. Defective films or leaks caused by faulty assembly are usually found when the cup is weighed on the third day. Continue to weigh the cup daily until the weight loss is constant to within 0.010 gram per day after two 24-hour periods. Then record daily weight loss for a continuous interval of 72 hours. The diffusion rate (D) in fluid ounces per square foot per 24 hours shall be the average of not less than three specimens when calculated from the following expression:

$$D = \frac{144 \text{ (average daily loss in grams)}}{\text{(Sp. Gr.) } (29.573)(3.122)(R^2)}$$

Where Sp. Gr. is the specific gravity of the test medium and R is the inside radius of the test cup. A diffusion rate of more than 0.12 fluid ounce/square foot per 24 hours shall constitute failure of this test.

4.5.2.11 Puncture resistance. FED-STD-191, method 5120 applies except that the ring clamp mechanism shall have an internal diameter of 3.00 inch, and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver, having a width of 0.312 ±0.010 inch and a thickness of 0.031 ±0.004 inch at the extreme tip. The piercing tip edges shall be rounded to a 0.010-inch radius. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees. The average of three test specimens shall be reported. Nonconformance to 3.3.5 and table II shall constitute failure of this test.

4.5.2.12 Low temperature crease resistance. Immerse three coated fabric specimens, each 8 inches square, in JP-5/JP-8 ST fuel conforming to note 1/ of table II, and condition for 24 hours, ±1 hour, at 73 ±5 °F. Remove specimens, place on a wire screen, and air dry with forced air movement at 73 ±5 °F, for 24 hours, ±1 hour. Then fold the specimens in half in each direction so that a folded corner occurs in the center of each specimen. Place each folded specimen under a 4-pound load and condition at -25 ±2 °F for 46 hours. At the end of the conditioning period, unfold the specimens while still at a

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temperature of -25 °F and examine visually. Signs of cracking, peeling, or delamination of any coating material shall constitute failure of this test. If the specimen does not fail, then subject the specimen to the diffusion test specified in 4.5.2.10 except position the specimen in the diffusion cup in such a manner that the center of the previously folded specimen coincides with the center of the cup. Nonconformance to 3.3.5 and table II shall constitute failure of this test.

4.5.2.13 Blocking. Place three coated fabric specimens 6 inches by 1 inch in an oven on a smooth surface in such a manner that the ends are overlapped 1 inch. Place a 4-pound weight directly on the overlapped area. After conditioning at a temperature of 158 ±2 °F, for 4 hours, remove the weight and take the specimens from the oven and condition for 1 hour at 73 ±5 °F and 65 ±2 percent humidity. Attach one end of the specimen in a suitable clamping device allowing the free end to hang down. Suspend a 4-ounce load from the free end of the specimens. Inability of the strips to separate within 5 seconds under the 4-ounce load shall constitute failure of this test.

4.5.2.14 Coating adhesion. Samples of coated fabric shall be bonded face to face to provide specimens for determining adhesion between the cloth and exterior coating(s); between the cloth and interior coating(s); between laminations of interior coatings and barrier (if used), and between laminations of exterior coatings. In forming this bond, the specimens shall be subjected to no heat or pressure other than that normally encountered in curing the coated fabric, except for minimal pressure necessary to ensure contact while the bond is setting. Samples shall undergo fuel extraction as described in 4.5.2.19 prior to coating adhesion test.

4.5.2.14.1 Test procedure. The adhesion shall be determined in accordance with ASTM D 413, machine method, except that the specimens shall be 2 inches wide. The specimens shall be of sufficient length to conduct adhesion test for both initial values and after fuel or water immersion. The adhesion results obtained on each immersed specimen shall be compared with the initial adhesion of the same specimen to determine percentage of adhesion retained. The reported adhesion and percent retention shall be the average of not less than two specimens. Attempts shall be made to cut the coating back to the cloth and to determine the adhesion value at the coating-to-cloth interface. However, if a specimen separates at a plane other than the bond of the coating to cloth (such as between layers of coating materials or between barrier film and coating) the adhesion value and the plane of failure shall be recorded. Immersed specimens shall be conditioned in the test fluid at 73 ±5 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the conditioning fluid. Immersion of specimens shall be in accordance with ASTM D 471. Nonconformance to table II shall constitute failure of this test. Any obvious bond failure evident after immersion but before stressing, even if the plane of failure is not sandwiched between the layers of fabric, shall constitute failure of this test.

4.5.2.15 Seam test. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, end closure, chafing patches, coated fabric flanges of fittings, etc.) shall be considered as seams and shall be subjected to all seam tests specified herein. The average breaking strength of five specimens for each type seam for each test will be reported for conformance to table III. Breaking strength specimens shall be 2 inches wide (parallel to the seam) and shall extend (perpendicular to the seam) 3 inches beyond both edges of the seam. No part of the test specimens shall be coated or covered during the fuel or water immersion periods. Specimens shall be cooled in the immersion fluids at 73 ±5 °F for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the immersion fluids. The average peel adhesion strength of two specimens for each type seam will be reported for conformance to table III. Peel adhesion specimens shall be of sufficient length to determine both the initial and after fuel or water adhesion values on the same specimen. If seam construction involves the use of binding thread, then the peel specimens

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shall be prepared with threads removed. Chafing patches shall be tested for peel only. Any nonconformance between reported values and the requirements of table III shall constitute failure of this test.

4.5.2.16 Dead load shear resistance. The test specimens shall be 1.0 inch ±.02-inch wide (parallel to the seam) and shall extend a minimum of 3 inches on each side of the seam. One index mark shall be scribed on each side of the seam to facilitate observation and measurement of slippage. Each specimen shall be subjected to the constant (dead load) tension force of 50 pounds ±0.50 pound at 180 ±2 °F. After 8 hours, examine each specimen while still under tension for signs of slippage or separation. Three specimens shall be tested for each determination. Slippage, by any specimen, greater than that specified in table III shall constitute failure of this test.

4.5.2.17 Fitting bond strength. Specimens shall be prepared by cutting through the flange such that parallel 1.0-inch wide sections are obtained from the straight portion of the oval fitting and 1.0-inch wedge shaped sections are obtained from the vent (or drain) and the curved portion of the oval fitting. The 1.0 inch shall be measured as a chord passing through the midpoint between the inside and outside diameters of the wedge shaped sections.

4.5.2.17.1 Initial bond strength. The coated fabric flanges shall be fastened together in one jaw of the test machine so that the jaw will be at least 1.0 inch from the nearest part of the aluminum. The aluminum flange shall be secured in the other jaw of the test machine and this jaw shall clamp only the aluminum and shall not compress the embedded part of the coated fabric flanges or clinch rings. The jaws shall be separated at a rate of 2.0-inch per minute at 75 ±5 °F and 65 ±2 percent humidity. The average of three test specimens shall be recorded as pounds per inch of width. An average value of less than 600 lbs/in shall constitute failure of this test.

4.5.2.17.2 Bond strength after fluid immersion. Three test specimens shall be immersed for the appropriate durations in each test fluid specified in table IV. No part of the specimens shall be covered or coated during immersion. Specimens from both the oval and vent/drain fittings shall be included in each test fluid. The test specimens shall be cooled in the immersion fluid to 75 ±5 °F for up to 60 minutes. The specimens shall be removed from the test fluid, one at a time and tested as in 4.5.2.17.1. Each test shall be completed within 3 minutes after removal from the test fluid. The average of three tests for each fluid shall be reported in pounds per inch of width. Nonconformance to the requirements in table IV shall constitute failure of this test.

4.5.2.17.3 Dead load shear resistance aluminum to fabric bond. Three specimens shall be clamped as in 4.5.2.17.1 and subjected to constant (dead load) tension force of 50 pounds at 180 ±5 °F. At the end of 8 hours, the specimens shall be examined for slippage or separation while under tension. Slippage in excess of 0.125 inch shall constitute failure of this test.

4.5.2.18 Peel adhesion of aluminum to coated fabric. Special test specimens shall be fabricated consisting of aluminum strips bonded to lengths of coated fabric. The aluminum strip shall be 12 inches long (minimum) by 2.00 inches ±0.05-inch wide by 0.125-inch thick (minimum) and shall be of the same alloy as that used in the aluminum fitting flanges. The coated fabric shall be 12 inches long (minimum) by 2.00 inches ±.05-inch wide and shall be of the same composition (and of the same state of cure before bonding) as that used in the coated fabric flanges. The coated fabric strip shall be uniformly bonded to the aluminum strip. The bond shall be formed using identical techniques and bonding agents used to bond tank fittings and shall be cured identically (time, pressure, temperature, etc.) to the process used in bonding tank fittings.

4.5.2.18.1 Test procedure. Specimens shall be tested as per ASTM D 429, method B. Three or more specimens shall be averaged for each fluid immersion

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test. The same identical specimens shall be used to determine the initial peel strength and the strength after fluid immersion and when computing the percentage of initial adhesion retained. Any nonconformance to the requirements of table IV shall constitute failure of this test.

4.5.2.19 Fuel extraction. Test specimens shall be prepared in accordance with the tests to be performed in tables I-IV and then submitted to the following extraction procedure:

a. Immerse the specimens fuel conforming requirements in the applicable table for 7 days at 160 ±2 °F.
b. Remove specimens from fuel and blot with paper towels.
c. Place specimens in a vacuum oven for drying 16 ±2 hours at 120 ±2 °F at 20 inches of mercury.
d. Samples will then be immersed in distilled water as required in tables I-IV.

4.5.2.20 Color. Color matching shall be in accordance with ASTM D 1729 for general match with color 33446 of FED-STD-595. Nonconformance to 3.11 shall constitute a failure of this test.

4.5.2.21 Coatings. Coatings properties shall be tested in accordance with the test methods shown in table I. Nonconformance to 3.3.4 or table I shall constitute failure of this test.

4.5.2.22 Coated fabric. Coated fabric properties shall be tested in accordance with test methods shown in table II. Nonconformance to 3.3.5 and table II shall constitute failure of this test.

4.5.2.23 Cloth. Cloth properties shall be tested in accordance with FED-STD-191, method 5804 and 5104. Properties shall apply after heat setting. In method 5804 alternate Corex D filters shall be removed. Specimens shall be raveled for method 5104 after 100 hours of accelerated weathering. The breaking strength warp and fill (lbs. [min.]) shall conform to FED-STD-191, method 5104. The edges of the test specimen shall be coated by dipping or brushing with a solution or other adhesive or sealed by melting with heat sufficiently to preclude yarn slipping while under test. Breaking strength after accelerated weathering less than 50 percent of the initial breaking strength shall constitute failure of this test. Both initial and aged breaking strength samples shall be run in warp and fill directions.

4.6 Inspection of packaging.

4.6.1 First article packaging inspection. The first article pack shall be inspected as follows:

4.6.1.1 Examination. The first article pack shall be examined for the defects listed in 4.6.2.2. The presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The first article pack level A and B shall be tested as follows:

4.6.1.2.1 Center of balance. The first article pack shall be lifted to a height of 36 ±4 inches at its lower end, with slings that have been centered over the marked center of balance and attached to the formed ends of the lifting bars. The lifted crate shall be examined for proper balance. Should the higher end of the crate exceed the height of the lower end by more than 24 inches, the first article pack shall be considered to have failed the test.

4.6.1.2.2 Slinging. The first article pack which has been lifted to the 36 inch height specified in 4.6.1.2.1, shall be rapidly raised to a height of 96 inches (±6 inches) at its lower end, abruptly stopped, and then rapidly lowered and rapidly stopped at the original height of 36 inches (±4 inches). This test shall be repeated 3 more times. Should there be any permanent

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deformations in the lifting devices, any crushing of the crate at a point where it contacts the lifting devices, any separation of lifting devices from the crate or any damage to the crate in any other location, including the first article pack shall be considered to have failed this test.

4.6.1.2.3 Unpacking. The first article pack shall have the sides, ends and top removed from its skid base in the manner and sequence prescribed by the contractor, and set aside. The first article pack shall have failed this test should the prescribed manner and sequence be ineffective, or should any of the contents fall free of the skid.

4.6.1.2.4 Repacking. As applicable (see 6.2), the crate body shall be reassembled to the crate skid base by matching the corresponding coded lines that have been marked on the crate components. Should the applied codes prove to be inaccurate, the first article package shall be considered to have failed the test.

4.6.2 Inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Examination. Each unit of product shall be examined for the following defects. The presence of one or more defects shall be cause for rejection.

No.	Defect	Paragraph Level A	Level B	Level C
123.	Fittings not removed from tank as specified.	5.2.1		
124	Removed hardware and accessories not preserved as specified.	5.2.1		
125	Hose assemblies not preserved as specified.	5.2.2		
126	Emergency repair items not preserved as specified.	5.2.3		
127	Technical publications, when required, not preserved as specified.	5.2.4		
128	Dust caps, plugs and cushioning material not applied as specified to the tank fittings.	5.2.5		
129.	Tank not dusted as specified.	5.2.5		
130.	Tank not folded and rolled as specified.	5.2.5		
131.	Eight deployment straps not as specified.	5.2.5		
132.	Deployment straps not positioned as specified.	5.2.5		
133.	Three tie-downs not as specified and the tank subjected to metal hazards as a result of the tie-downs.	5.2.5		
134.	Three lifting slings not as specified.	5.2.5		
135.	Lifting slings not positioned and secured as specified.	5.2.5		
136.	Wood/Plywood not treated as specified.	5.3.1.1		

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No.	Defect	Paragraph Level A	Level B	Level C
137.	Fasteners not galvanized as required.	5.3.1.2		
138.	Metal surfaces of the product that contact the treated wood/plywood not protected with barrier material as specified.	5.3.1.3		
139.	Shipping container not the MIL-C-104 type, class or style specified.	5.3.1.4	5.3.2	
140.	Skid base not as specified.	5.3.1.4.1	5.3.2	
141.	Crate top not as specified.	5.3.1.4.2	5.3.2	
142.	Crate sides not as specified.	5.3.1.4.3	5.3.2	
143.	Crates ends not as specified.	5.3.1.4.4	5.3.2	
144.	Filled consolidation container not secured to skid base as required.	5.3.1.4.5	5.3.2	
145.	Interior of container not lined to prevent snagging as specified.	5.3.1.4.6	5.3.2	
146.	Interior of container not lined to prevent abrasion as specified.	5.3.1.4.7	5.3.2	
147.	Hose assemblies not secured within the crate as specified.	5.3.1.4.8	5.3.2	
148.	Lifting devices not as specified.	5.3.1.4.9	5.3.2	
149.	Lifting bar not as specified.	5.3.1.4.9.1	5.3.2	
150.	Lifting bar adapter not as specified.	5.3.1.4.9.2	5.3.2	
151.	Lifting device fasteners not as specified.	5.3.1.4.9.3	5.3.2	
152.	Reinforcing joists for lifting devices not as specified.	5.3.1.4.9.4	5.3.2	
153.	Outside dimension of crate exceed maximum acceptable sizes.	5.3.1.4.10	5.3.2	
154.	Repair items, technical publications, fittings and accessories not packed and cushioned as specified.	5.3.1.5	5.3.2	
155.	Sheathing upgrade, when required, not as specified.	5.3.1.6	5.3.1.6	
156.	Skid base and cover for level C packing not as specified.			5.3.3.1
157.	Headers not as specified for level C.			5.3.3.1.1
158.	Crate size for level C not as specified.			5.3.3.1.2
159.	Emergency repair items, technical publications, fittings, hose assemblies and accessories not packed in a manner to provide protection from damage and deterioration for level C.			5.3.3.2

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No.	Defect	Paragraph Level A	Level B	Level C
160.	MIL-STD-129 marking not as specified.	5.4	5.4	

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and tests within the time frame specified (see 6.2), to prove, prior to starting production packaging, that the applied preservation, packing and marking comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.7). The first article pack may be accomplished utilizing either the first article model or production model. If the first article model is utilized, and the Government requests a comparison between the first article model and production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 Preservation. Each complete collapsible tank shall be preserved for level A as follows:

5.2.1 Fittings and accessories. Fittings shall be removed from the tank at the quick-disconnect. The removed hardware, the gate valve assembly, and ball valve assembly (see 3.6) shall be wrapped with cushioning material conforming to PPP-C-795, A-A-1898, type I, class B, or PPP-C-1797, secured in place with tape conforming to PPP-T-60, type IV or MIL-T-22085, type II.

5.2.2 Hose assemblies. The drain hose assembly and the 6-inch hose assemblies (see 3.6) shall be sealed with their respective caps and plugs to prevent entry of foreign matter.

5.2.3 Emergency repair items. Emergency repair items of like description shall be preserved together in accordance with MIL-P-116, method IC-1 or IC-3. Gasket and O-rings shall be preserved in a manner as to prevent damage or deformation.

5.2.4 Technical publications. Technical publications when required shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.5 Tank. All dust caps and plugs shall be installed and the permanently attached fittings shall be covered with cushioning material conforming to PPP-C-795, A-A-1898, type I, class B, or PPP-C-1797, secured in place with tape conforming to PPP-T-60, type IV or MIL-T-22085, type II. The tank interior and exterior surfaces shall be dusted with talc or other anti-sticking compound to prevent adhesion to itself. The tank shall be laid flat and completely collapsed, folded and rolled, and then secured as specified in figures 11, 12, 13, 14 and 15. Eight deployment straps (see 3.6) shall be attached to the handles and rolled up with the tank to assist in the tank's deployment. Additionally, three polyester lifting slings (see 3.6) shall be applied around the rolled up tank to assist in lifting. Slings shall be secured in place without damage to the tank. The tank shall be secured to the skid using three tie-down straps. Each tie-down strap shall be webbing, a minimum of 2 inches in width, 20 feet in length, have a tensile strength of 5,000 pounds and with a hand operated ratchet-winch attached on one end. The tie-down straps shall be wrapped with cushioning material to prevent chafing or possible contact with the tank fabric. The tank shall not be subjected to the hazards of any sharp metal objects used to secure the tank to the skid or to remove the tank from the skid.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.3.1 Level A. When level A packing is specified, the following special considerations shall be afforded this product because of an anticipated extended period of storage under unfavorable environmental conditions.

5.3.1.1 Wood preservation. All lumber and plywood used in the construction of the exterior crates/boxes shall be treated with one of the following wood preservative solutions: 2% copper naphthenate, 3% zinc naphthenate, or 1.8% copper 8 quinolinolate.

5.3.1.2 Fasteners. All fasteners used in the fabrication of any component of a wooden container, all fasteners used in the blocking, bracing, and anchoring of the contents to any component of a wooden container, and all fasteners used to assemble the components of a wooden container together, shall be galvanized fasteners.

5.3.1.3 Surface protection. All metal surfaces of the product that come in direct contact with the treated wood/plywood, shall be protected with non-corrosive barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall completely cover the contact surfaces and shall be secured in place when required, with tape conforming to MIL-T-22085, type II, or PPP-T-60, type IV.

5.3.1.4 Exterior container. Each tank and the hose assemblies shall be packed in a crate conforming to MIL-C-104, type II, class 2, style A, except the skid base shall be as follows:

5.3.1.4.1 Skid base.

5.3.1.4.1.1 Headers. The 4 x 4 headers shall be as specified in MIL-C-104 except that the holes for the carriage bolts shall be counterbored to the depth necessary to prevent any portion of the fastener extending inside the crate.

5.3.1.4.1.2 Load-bearing floorboards. The load-bearing floorboards shall be of the size specified for the headers and the holes for the carriage bolts shall be counterbored in the same manner. Additionally, in order to provide a smooth, flat surface for the rolled tank to rest on, an upper plywood deck shall be applied on the top of the headers and load-bearing members of the skid base. The upper plywood deck shall be 0.75 inch thick and additional supporting members shall be positioned between and parallel with the headers and load-bearing members, as applicable, to minimize the deflection of the deck. The load-bearing floorboards, and additional supporting members as required, shall be attached to the skids with either two lengths of carriage bolts. The shorter length shall attach the members to the skids only. The longer length shall attach the members to the skids and rubbing strips with a minimum of 3 fasteners extending through the two longer rubbing strips. The short, 16 inch long, center rubbing strip shall be attached to the skid base with a total of 2 fasteners of either or both lengths.

5.3.1.4.1.3 Rubbing strips. The rubbing strips shall be nominal 4-inch thick lumber of the same width as the skids and shall be beveled full depth at an angle of 45 degrees at sling and forklift-truck openings. The rubbing strips shall be positioned on each skid to provide two 14-inch wide openings for forklift-truck access. Such openings shall be spaced 30 inches apart, center to center, and shall be positioned to straddle the center of balance of the loaded crate. The sling openings at the ends of each skid shall be 8 inches in length. Attachment of rubbing strips to the skid base shall be as specified in 5.3.1.4.1.2; however, no carriage bolt shall be positioned closer than 5 inches to the end of the top (longer) side of the rubbing strip.

5.3.1.4.2 Top. The crate top shall be lumber sheathed in accordance with MIL-C-104, except 0.50-inch thick plywood shall be used for the inside sheathing. Additional joists shall be required in each top to compliment the lifting device feature (see 5.3.1.4.9.4).

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5.3.1.4.3 Sides. The crate sides shall be as specified in MIL-C-104 except that the plywood sheathing shall be 0.50-inch thick plywood, the lower frame members shall be nominal 2-inch thick lumber, and the side struts shall be nominal 2-inch thick lumber. (The side end struts shall be nominal 4 x 4 as specified in MIL-C-104.)

5.3.1.4.4 Ends. The crate ends shall be as specified in MIL-C-104 except that the plywood sheathing shall be 0.50-inch thick plywood.

5.3.1.4.5 Filled consolidation container(s). The filled consolidation container(s) (see 5.3.1.5) shall be positioned on the skid base in a manner to prevent movement. The method of attachment shall utilize the skid base only; it shall not be dependent upon the application of devices to the sides, top or end(s) of the crate.

5.3.1.4.6 Anti-snagging provisions. To prevent the tank from snagging against interior framing and fasteners, all applicable interior surfaces of the crate shall be lined with either plywood, hardboard or waferboard and secured in place by gluing or nailing. Nails, if used, shall be the minimum necessary to secure the panels and shall be placed in locations so that loosening of the nails shall not create a snagging hazard. Those surfaces of consolidation container(s) with exposed cleats that may cause snagging shall be lined in the same manner.

5.3.1.4.7 Anti-abrasion provisions. To prevent tank abrasion, all interior surfaces of the crate that will contact the tank, including any applicable outer surface of the consolidation container(s), shall be lined with a layer of scrap tank fabric, cushioning material conforming to PPP-C-1797, or polyethylene film conforming to L-P-378, type I, class 1, with a minimum thickness of 6 mils. The material shall be applied in such a manner that it does not interfere with the removal of the crate top, sides and ends as a joined one-piece assembly or as individual panels. The means of attaching the material shall not create a hazard within the crate that could possibly damage the tank.

5.3.1.4.8 Hose assemblies. The four 10-foot-length hose assemblies, preserved as specified in 5.2.2, shall be positioned and secured within the crate in such a manner to assure that when the crate top, sides and ends are removed from the skid base, the hoses remain secured in place without damage to the tank.

5.3.1.4.9 Lifting devices. Lifting devices shall be applied to the sides of the crate to assist in slinging operations and unpacking operations. The devices shall be applied to the crate in the manner shown in figure 16 and shall be constructed in accordance with the applicable figure referenced in the appendix to MIL-C-104, except for the following:

a. Each lifting device shall be a two-piece assembly, consisting of a lifting bar (see 5.3.1.4.9.1) and a lifting bar adapter (see 5.3.1.4.9.2).
b. The formation for attaching a sling to the upper portion of the lifting bar shall be formed with a 0.5000-inch (±0.0625 inch) inside radius.
c. The devices, when assembled together, shall provide the necessary strength to withstand the lifting forces specified in MIL-STD-209 for equipment weighing less than 20,000 pounds. The lifting devices shall be tested in accordance with MIL-STD-209. Inability to meet the requirements of MIL-STD-209 shall constitute failure of this test.

5.3.1.4.9.1 Lifting bar. The lifting bar shall be constructed of 2.50-inch wide x 0.375-inch thick, galvanized, structural steel, bar stock with a tensile strength to withstand the lifting forces specified in 5.3.1.4.9. One end of the lifting bar shall be formed to provide the means by which lifting slings may be utilized (see 5.3.1.4.9). The opposite end of the lifting bar shall be provided with two tapped holes and one untapped hole. The two tapped holes provide the means by which the lifting bar adapter is assembled to the

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lifting bar (see figure 17). Clearance holes shall be provided in the plywood sheathing of the crate sides to accept the threaded portion of the fastener that protrudes beyond the lifting bar. The untapped hole, which is the middle hole (see figure 17) provides the means by which the assembled device is attached to the skid of the base. The lifting bar shall be provided with 3 additional holes along its length for the purpose of attaching the lifting bar to the assembled side panel of the crate (see figure 16 and the applicable figure referenced in the appendix to MIL-C-104). The outermost of these holes shall be positioned one each in the upper and lower frame members of the crate side. The innermost hole shall be centered on the lifting bar, equidistant between the outermost holes, utilizing a side strut for inside support. When a side strut is positioned so that such centered innermost hole would fall on or near the butt joint of the plywood sheathing or near the edge of the side strut, the hole may be positioned a maximum of 6 inches off center providing such location results in the center line of the hole being no closer than 0.625-inch to the edge of the metal lifting bar or 1.125 inches away from the edge of the plywood sheathing or side strut. When no side strut falls within these limitations, a 10-inch long wood reinforcing batten of the same thickness and width as the side strut, shall be attached to the inside of the plywood sheathing in the manner specified for the struts in MIL-C-104. The batten shall be parallel with the lifting bar and shall be positioned so that the hole will be centered in the batten. Prior to galvanizing, the lifting bar surfaces shall be made free of burrs and sharp edges and the corners shall be rounded.

5.3.1.4.9.2 Lifting bar adapter. The lifting bar adapter shall be constructed of 2.50-inch wide x 0.375-inch thick, galvanized, structural steel, bar stock with a tensile strength to withstand the lifting forces specified in 5.3.1.4.9. The adapter shall be formed at a 90-degree angle which has been offset 75 degrees on each leg to correspond with the placement angle of the lifting bars (see figures 16 and 17). Each leg of the adapter shall be provided with 3 holes in the manner shown in figure 17, so that it may be used in the left hand location or the right hand location by simply selecting the applicable leg which has been embossed/stamped as either "Right" or "Left" (see figure 17). Prior to galvanizing, the lifting bar adapter surfaces shall be made free of burrs and sharp edges and the corners shall be rounded.

Special note: The lifting bar can be positioned either left or right on the crate sides. The holes in the lifting bar adapter are to match those hole locations so that it too can be positioned either left or right by placing the proper leg in the vertical position.

5.3.1.4.9.3 Lifting device fasteners. The bolts used to assemble the lifting bar and lifting bar adapter together, and the bolts used to attach the lifting device to the crate side panel and skid base, shall be of such size and material to provide the necessary strength to pass the lifting forces specified in 5.3.1.4.9. Each hole for attaching the lifting devices to the crate side panels shall be appropriately counterbored so that no portion of the fastener extends inside the crate.

5.3.1.4.9.4 Reinforcing joists for lifting device. When no top joists coincide with the positioning of the lifting bars on either or both sides of the center of balance of the loaded crate (see figure 16), additional reinforcing joists shall be placed at those positions to distribute the load when the crate is lifted with slings. The additional reinforcing joist(s) shall conform to the size requirements of MIL-C-104 and shall be positioned to offer maximum support to the lifting force.

5.3.1.4.10 Crate size. The maximum outside dimensions of the crate shall be 19 feet 2 inches in length, 72 inches in width, and 48 inches in height, except that when the tank is to be required for multiple training or other deployment type exercises with repeated unpacking and repacking (see 6.2 and 6.8), the outside height dimension shall be increased to 58 inches.

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5.3.1.5 Emergency repair items, technical publications, fittings and accessories. The preserved emergency repair item (see 3.7 and 5.2.3), the technical publications when required (see 5.2.4), the removed fittings (see 5.2.1), and the accessories, except for the four 10-foot length hoses (see 3.6), the lifting straps and the lifting slings (see 3.6 and 5.2.5), shall be packed in one or more close-fitting, wood cleated, plywood boxes. The box(es) shall comply with PPP-B-601, type overseas, style A or B. Cushioning, blocking and bracing shall be applied to the contents in accordance with MIL-STD-1186 to prevent damage. The cushioning material specified in 5.2.1 herein may be used in lieu of or in conjunction with the cushioning material prescribed in MIL-STD-1186. Strapping shall be zinc-coated and in accordance with the appendix to the box specification.

5.3.1.6 Sheathing upgrade. When specified (see 6.2), the plywood sheathing specified in 5.3.1.4.3 and 5.3.1.4.4 for the crate sides and ends shall be increased in thickness to a total of 0.75 of an inch. This option shall be exercised only when a quantity of tanks are known to be required for multiple training or other deployment type exercises where the tanks must be repeatedly unpacked and repacked (see 6.8).

5.3.2 Level B. Each complete tank, including accessories, emergency repair items and technical publications, when required, preserved as specified in 5.2, shall be packed for level B the same as specified in 5.3.1 for level A except that the special considerations specified in 5.3.1.1, 5.3.1.2 and 5.3.1.3 shall not be required. These special considerations are deleted because it is anticipated that this product will not be subjected to an extended period of storage under unfavorable environmental conditions.

5.3.3 Level C (Sierra design). When level C packing is specified, the following special considerations shall be afforded this product because of anticipated storage in International Standard Organization (ISO) containers at depot by the Army.

5.3.3.1 Skid and cover. The skid for packing the tank shall conform to figure 18. The cover shall conform to figure 19 and may be a throw-away design utilized to provide physical protection for the tank and allow for two high stacking during transport from the contractor to the initial destination. After the tank is secured to the skid, the cover shall be attached to the skid using lag bolts as shown in figure 19.

5.3.3.1.1 Headers. The header shall be 2-3/8 inches (actual) x 12 inches x 60 inches and the holes for the carriage bolts shall be countersunk to the depth necessary to prevent any portion of the fastener extending inside the crate (see figure 18).

5.3.3.1.2 Crate size. The maximum outside dimensions of the crate shall be 16 feet 6 inches in length, and 5 feet 9 inches in width, and 4 feet in height.

5.3.3.2 Emergency repair items, technical publications, fittings, hose assemblies and accessories. The emergency repair items, technical publications, when required, removed fittings, hose assemblies and the accessories shall be packed in such a manner to provide protection from deterioration and damage from handling, shipment and any storage from the contractor to the initial destination.

5.4 Marking. In addition to any special or identification markings required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129, including all appendices, and as applicable to overseas shipments. Special care must be exercised to assure the following:

a. That interior and exterior packing lists are applied as specified.
b. That marking surfaces are prepared as specified.

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c. That all markings are waterproofed by coating the entire surface of the label with a waterproof varnish, clear acrylic coating compound, or label adhesive as specified.
d. That, the date of manufacture and serial number of the tank is stenciled on the exterior of the crate and is also stenciled on the skid. These markings must be accessible without having to unroll the tank.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The collapsible tank is intended for use as a fuel storage container for diesel and jet fuels when large capacity quick storage facilities are needed, and where permanent fuel storage facilities are not available. It is intended that the collapsible tank provide a ten-year shelf life (extendable) and two-year use life (extendable).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame and number of tanks required for submission of first article (see 3.2).
d. When the Government will conduct any or all of the first article model examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the contractor (see 3.2).
e. Time frame for submission of process data (see 3.2.2).
f. When fitting assemblies are not specified (see 3.4.4.2).
g. When accessories for each tank are not as specified (see 3.6).
h. When repacking is required (see 4.6.1.2.4 and 5.3.1.4.10).
i. Time frame for submission of first article pack (see 5.1).
j. Level of packing required (see 5.3).
k. When sheathing upgrade is required (see 5.3.1.6 and 6.8).
l. Any special marking (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.2.1.1	DI-4901	First article inspection procedure.
3.2.1.1	DI-4902	First article inspection report.
3.2.2	DI-E-30133	Process specification.

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.4 Nylon yarn. The nylon material Elvamide 8061, manufacturers' code 18873, was found to meet the performance requirements and is suggested for use as a guide only. The Government assumes no responsibility for material.

6.5 First article. When a first article inspection is required, the items should be a preproduction model and first article test samples (3.2.1). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Subject term (key word) listing.

Containers, collapsible, fuel
Tanks, collapsible, fuel
Tank, fabric, collapsible, fuel

6.7 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor's obligation to preserve, pack and mark the tanks in accordance with this specification.

6.8 Sheathing upgrade. When a quantity of tanks are known to be required for multiple training or other deployment type exercises where the tanks must be repeatedly unpacked and repacked, the procuring activity should exercise the option of upgrading the plywood sheathing for the sides and ends (see 5.3.1.6).

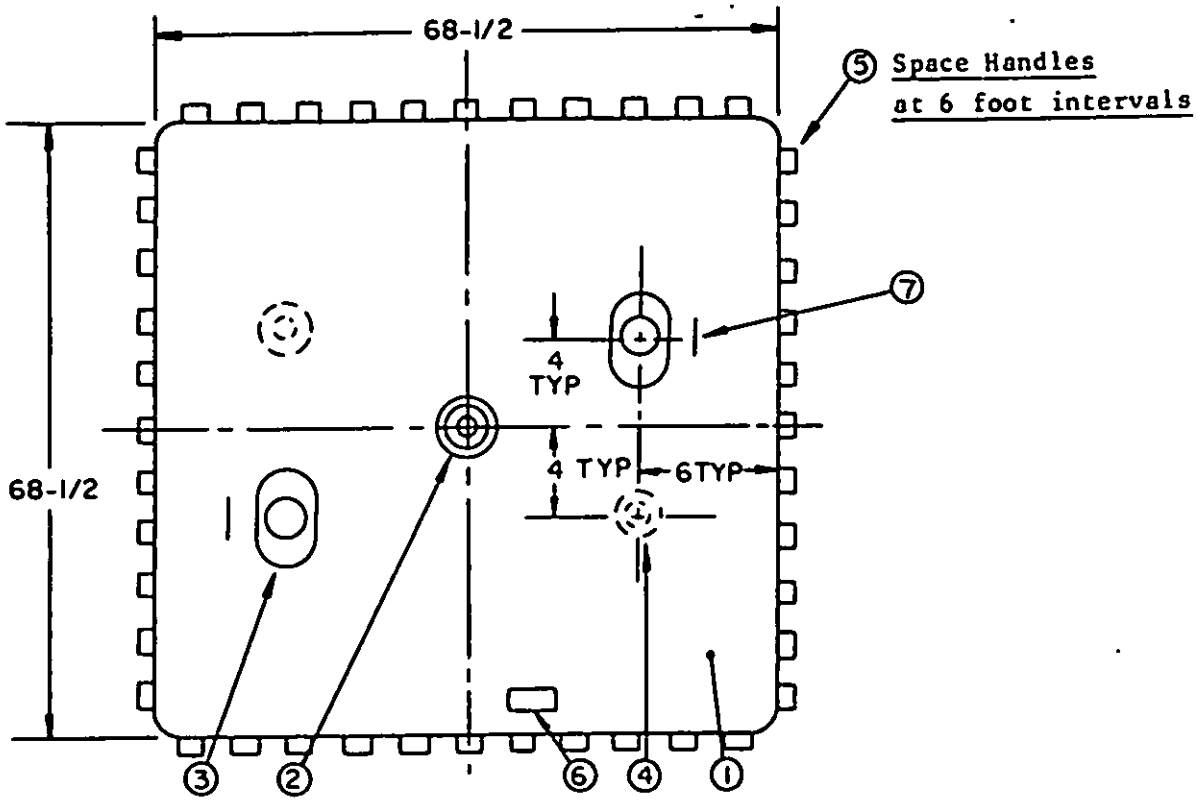
6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - YD1
Air Force - 99

Review activities:
Navy - MC
Air Force - 82
DLA - CS

Preparing activity:
Army - ME
Project 5430-0231

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REQUIRED DIMENSIONS OF EMPTY TANK

NOTES

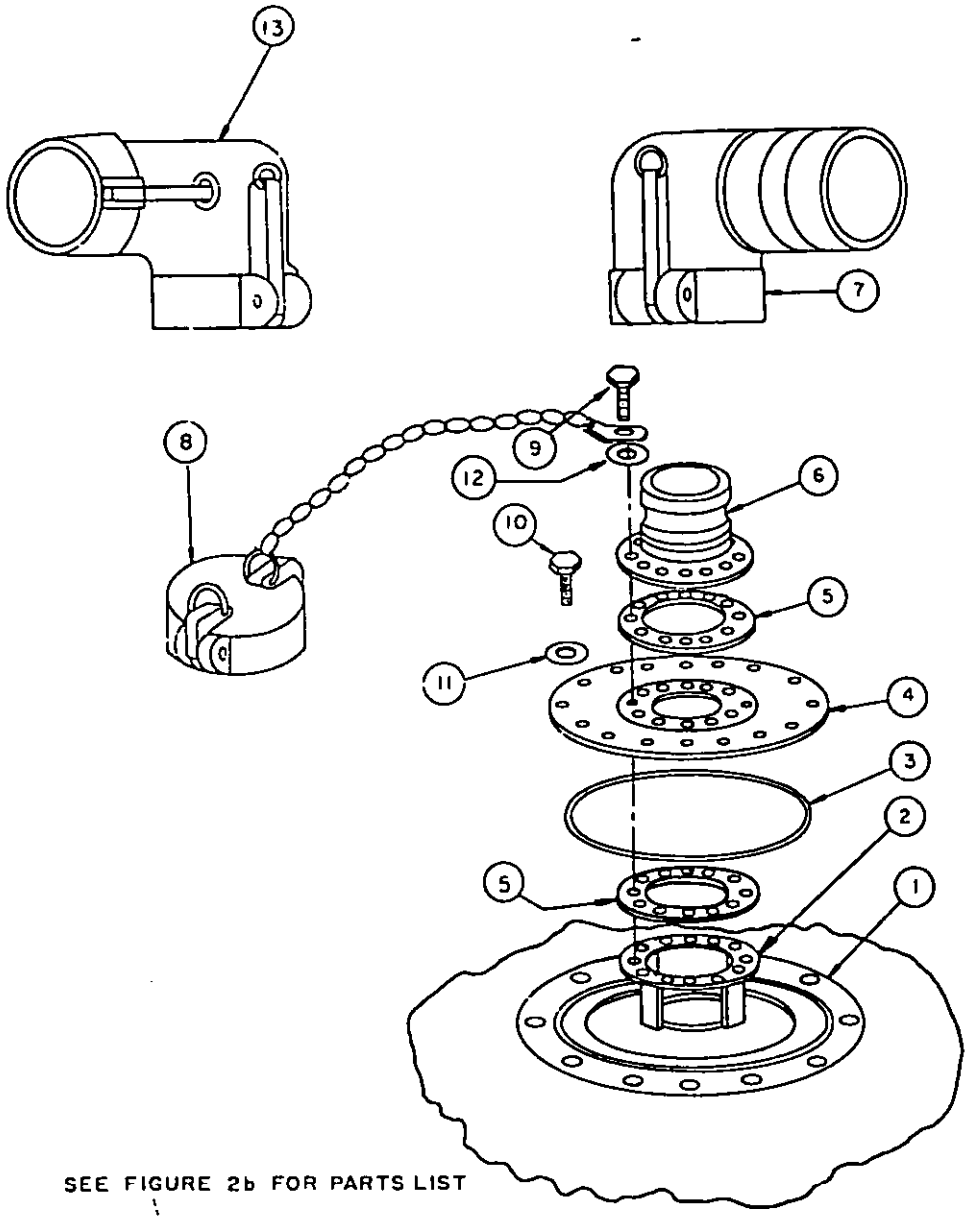
1. ALL DIMENSIONS IN FEET
2. TOLERANCES: FITTING LOCATION ± 2 FEET. OTHERS ± 1 FOOT
3. SHAPE OF CORNERS OPTIONAL
4. THE FOLLOWING SIZE CHAFING PATCHES SHALL BE BONDED INSIDE AND OUTSIDE THE TANK OPPOSITE EACH FITTING:
   2 VENT - 1-1/2 FT. DIA, ±1 INCH OR SQUARE
   3 FILL - DISCH - 3 FT. DIA, ±1 INCH OR SQUARE
   4 DRAIN - 1-1/2 FT. DIA. ±1 INCH OR SQUARE

ITEM	QTY	DESCRIPTION
①	1	TANK, 5000 BARREL
②	1	VENT ASSY
③	2	FILLER-DISCHARGE ASSY
④	2	DRAIN ASSY
⑤	44	HANDLE (11 PER SIDE)
⑥	1	LABEL
⑦	2	WARNING NOTICE

FIGURE 1. Tank, fabric collapsible, 5000 barrel fuel

X-4124B

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SEE FIGURE 2b FOR PARTS LIST

FIGURE 2a. Filler/discharge assembly.

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FIND NO.	PART NO. OR IDENT NO.	QTY FILLER FITTING ASSY	QTY DISCHARGE FITTING ASSY	DESCRIPTION	SPECIFICATION	MATERIAL
1	FIGURE 3	1	1	ACCESS DOOR FITTING-COMPRESSION TYPE		
2	FIGURE 4	1	1	SUCTION STUB		AL ALY, ANODIZED
3	MS9021-383	1	1	"O" RING		RUBBER
4	FIGURE 9	1	1	OVAL CLOSURE PLATE-COMPRESSION FITTING		AL ALY, ANODIZED
5	FIGURE 10	2	2	GASKET		BUNA N RUBBER
6	MS27023-19	1	1	COUPLING HALF, FLANGED, 6 IN. SIZE		AL ALY, ANODIZED
7	81718/633 K-6	-	1	ELBOW, FEMALE TO MALE, 6 IN. X 90° SIZE	OPW OR EQUAL	AL ALY, ANODIZED
8	MS27028-19	1	1	DUST CAP, W/ 1200 L CHAIN		AL ALY, CAP, ANODIZED
9		12	12	BOLT, HEX HD, GR 8 .375-24 UNF-2A X 1.50 L	ANSI B18.2.1	STL, ZINC COATED .003 THK
10		20	20	BOLT, HEX HD, GR 8 .250-20 UNC-2A X 1.00 L	ANSI B18.2.1	STL, ZINC COATED .003 THK
11		20	20	WASHER, PLAIN, .281 ID	ANSI B18.22.1	STL, ZINC COATED .003 THK
12		12	12	WASHER, PLAIN, .406 ID	ANSI B18.22.1	STL, ZINC COATED .003 THK
13	81718/633 K8-6	1	-	ELBOW, FEMALE TO FEMALE, 6 IN. X 90° SIZE	OPW OR EQUAL	AL ALY, ANODIZED

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. LUBRICATE O-RING AND GASKET, FIND NO. 3 AND 5, WITH SILICONE COMPOUND BEFORE ASSEMBLING HARDWARE.
3. ZINC COATING SHALL BE IN ACCORDANCE WITH ASTM A 153, CLASS D.

FIGURE 2b. Filler/discharge assembly parts list.

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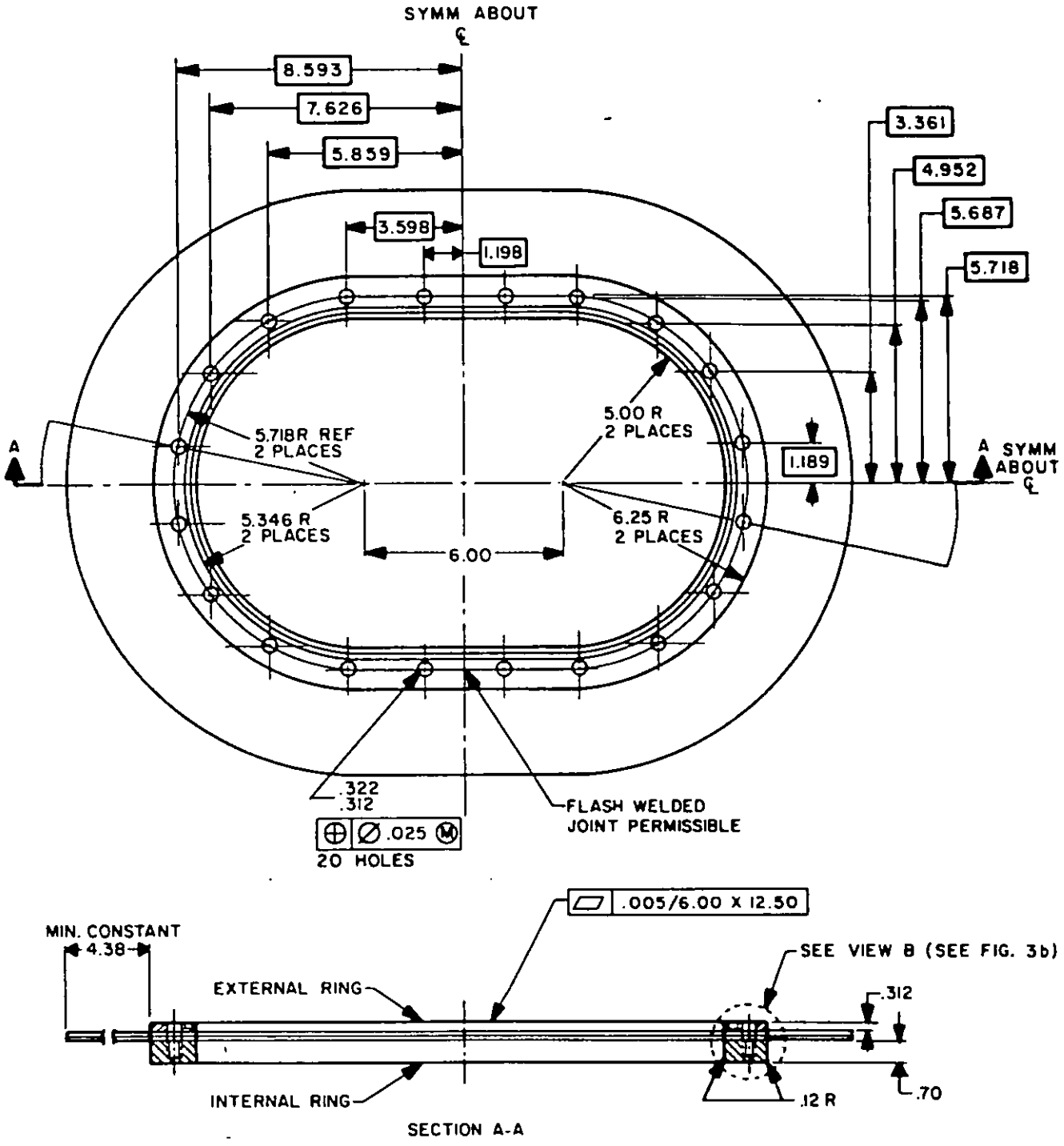
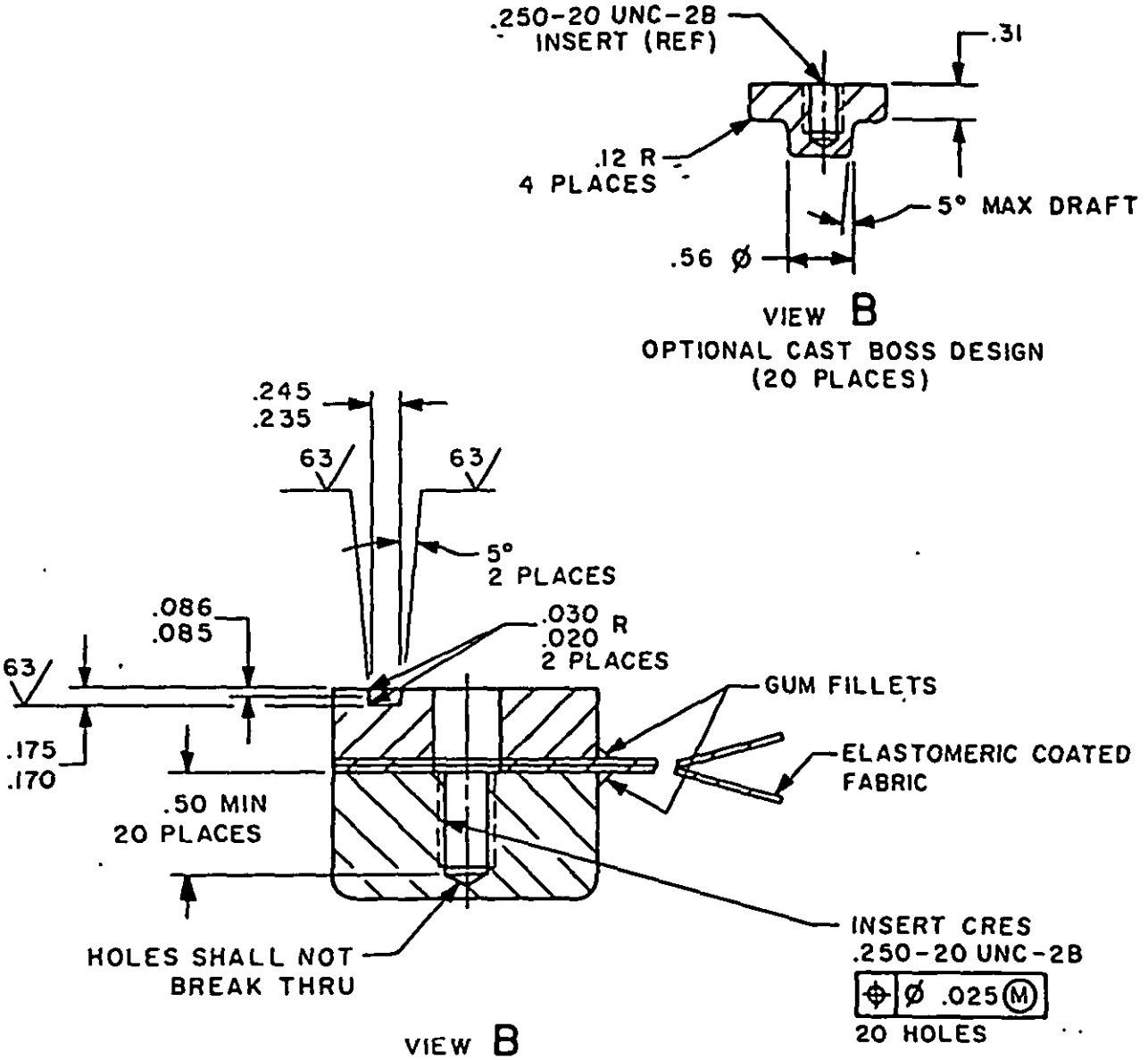


FIGURE 3a. Access door fitting-compression type.

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NOTES:

1. DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX = ± .06
   .XXX = ± .015
   ANGLES = ± 2°
2. ALUMINUM PARTS SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II, CLASS 2.
3. ADHESIVE BOND FABRIC AND RINGS.

FIGURE 3b. Access door compression type fittings.

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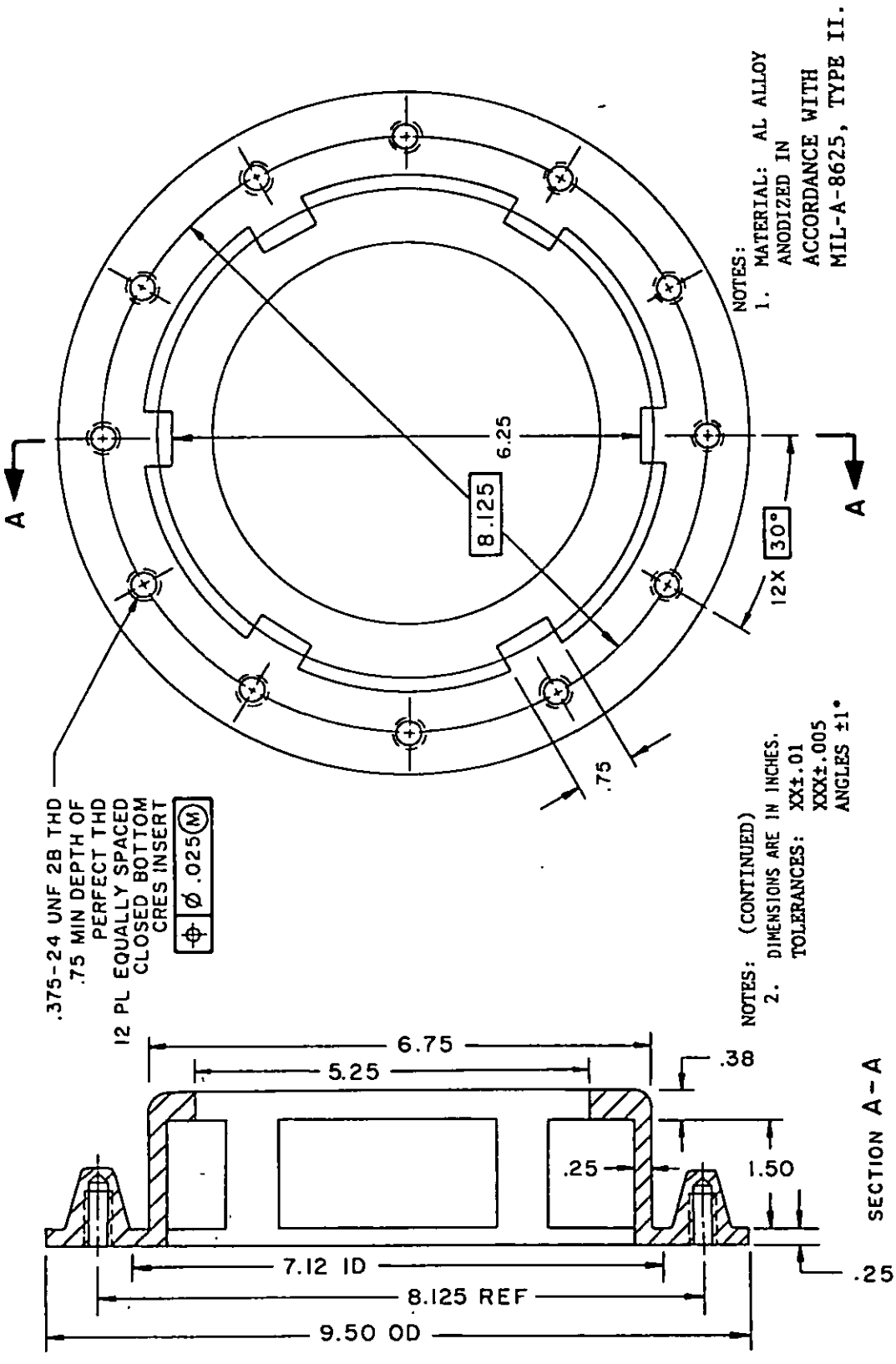


FIGURE 4. Suction stub for filler fitting assembly.

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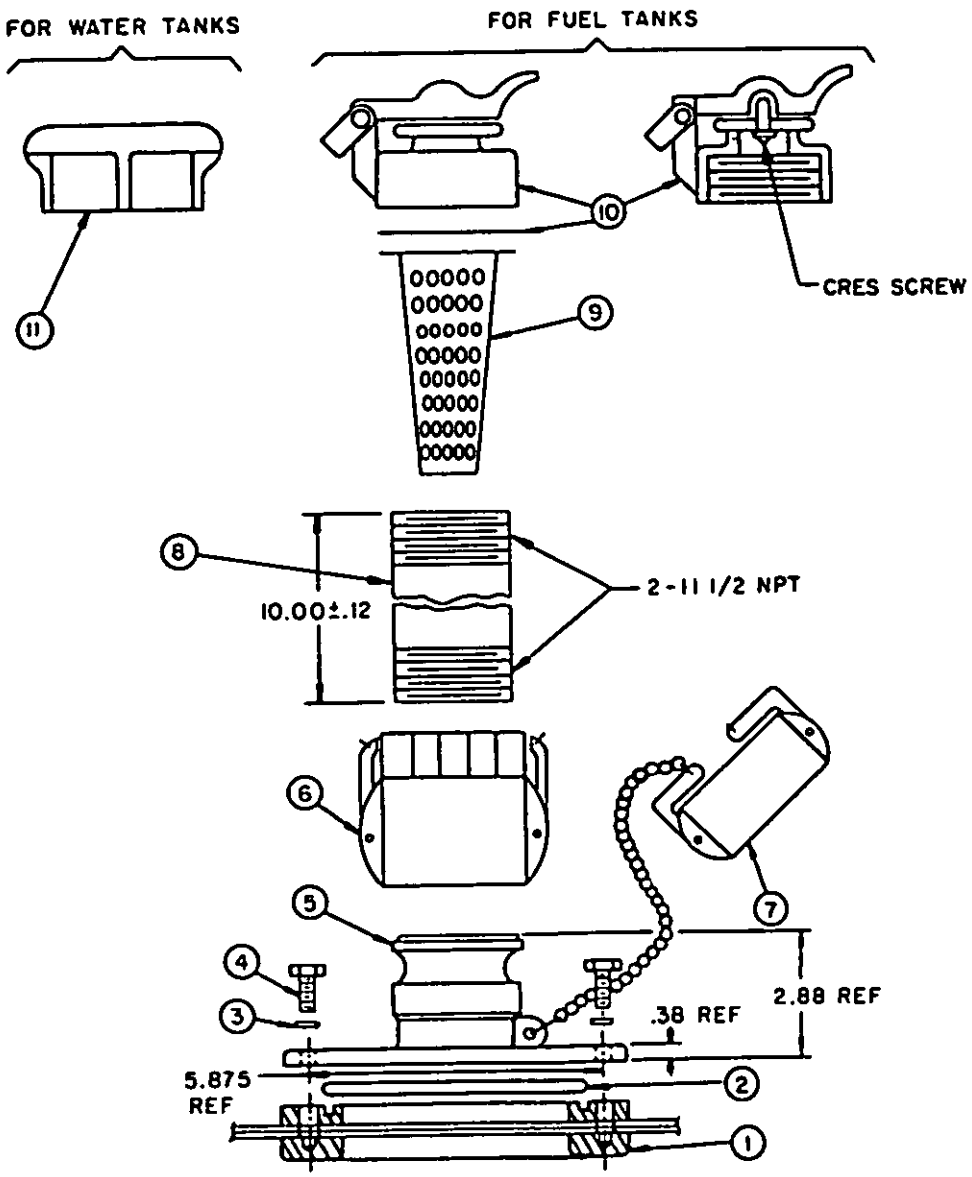


FIGURE 5a. Vent fitting assembly.

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NOTES:

1. LUBRICATE O-RING, FIND NO. 2 WITH SILICONE COMPOUND BEFORE ASSEMBLING HARDWARE.
2. UNLESS OTHERWISE SPECIFIED, ANODIZED ALUMINUM SHALL BE IN ACCORDANCE WITH MIL-A-8625, TYPE II, CLASS 2.
3. ZINC COATING SHALL BE IN ACCORDANCE WITH ASTM A153, CLASS D.
4. DIMENSIONS ARE IN INCHES.
5. CONNECT DUST CAP KEEPER CHAIN TO INTEGRAL ATTACHMENT POINT.
6. FOR WATER TANKS DELETE FIND NO. 9 AND 10. REPLACE WITH FIND NO. 11.

FIND NO.	PART NO. OR IDENT NO.	QTY	DESCRIPTION	SPECIFICATION	
1.	FIGURE 6	1	VENT AND DRAIN FITTING-COMPRESSION TYPE		
2.	MS29513-250	1	"O" RING GASKET		SYN RUBBER
3.		8	WASHER, PLAIN, .281 ID	ANSI B18.22.1	STL, ZINC COATED, .003 THK
4.	B1821BH025 C100N	8	BOLT, HEX HD, GR 8 .250-20 UNC-2A X 1.0 L	ANSI B18.2.1	
5.	MS27023-21	1	COUPLING HALF, QUICK DISCONNECT CAMLOCKING TYPE, MALE FLANGED		AL ALLOY, ANODIZED
6.	MS27024-11	1	COUPLING HALF, QUICK DISCONNECT CAMLOCKING TYPE, FEMALE, THREADED		AL ALLOY, ANODIZED
7.	MS27028-11	1	DUST CAP, 2" WITH 12 INCH LG SECURITY CHAIN STYLE B		AL ALLOY, ANODIZED
8.		1	PIPE, 2 INCH NOM, SCHEDULE 40		AL ALLOY, ANODIZED
9.		1	FLAME ARRESTOR		AL ALLOY ANODIZED
10.		1	RELIEF CAP WITH GASKET	PROTECTO SEAL # EX1333-2" OR EQUAL	AL ALLOY, ANODIZED
11.	72423/6170	1	MUSHROOM TYPE VENT WITH SCREEN NO. 30 MESH - SIZE 2 IN. NPT		BODY-IRON, GALV SCREEN-BRASS

FIGURE 5b. Vent fitting assembly parts list.

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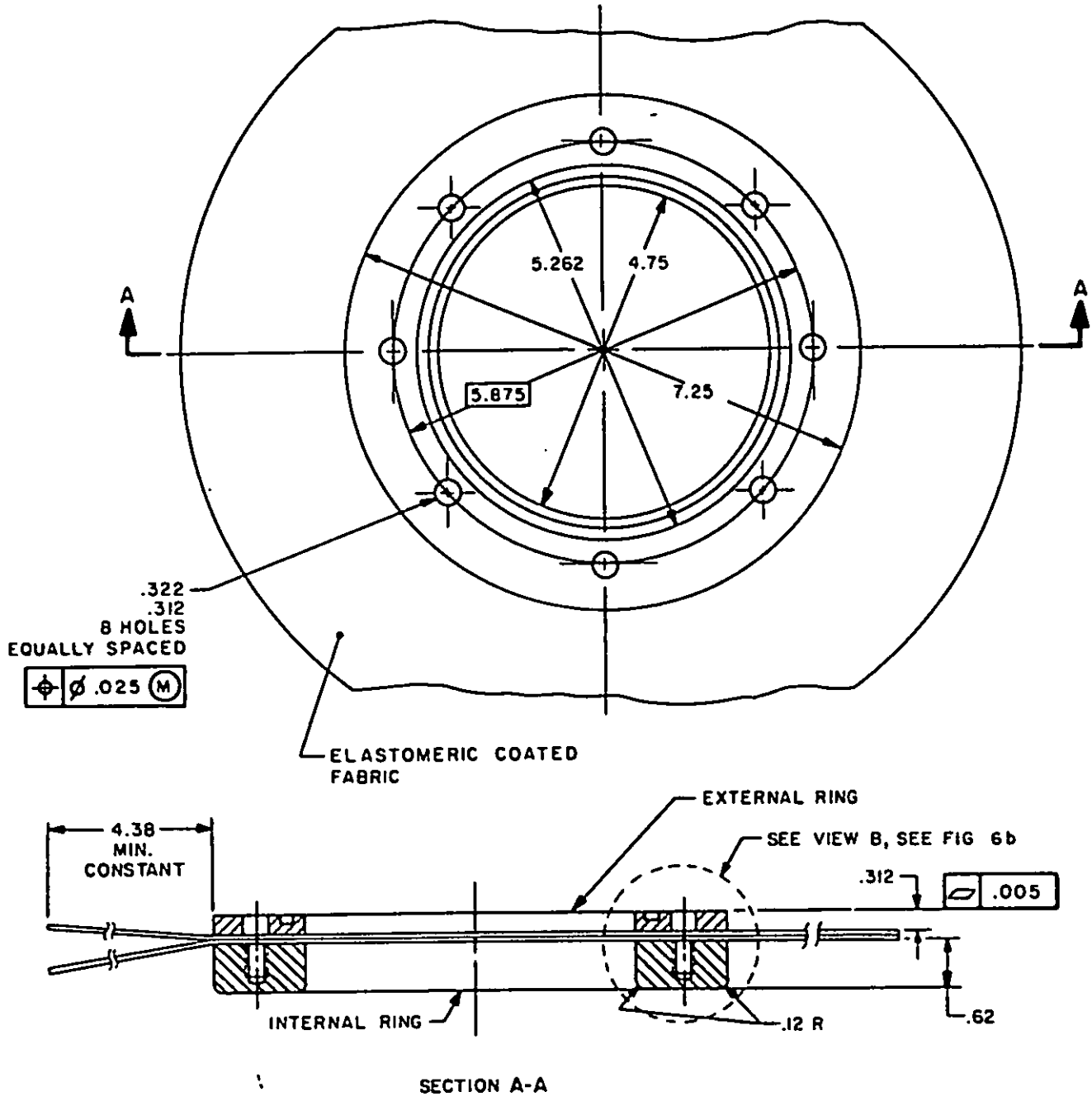
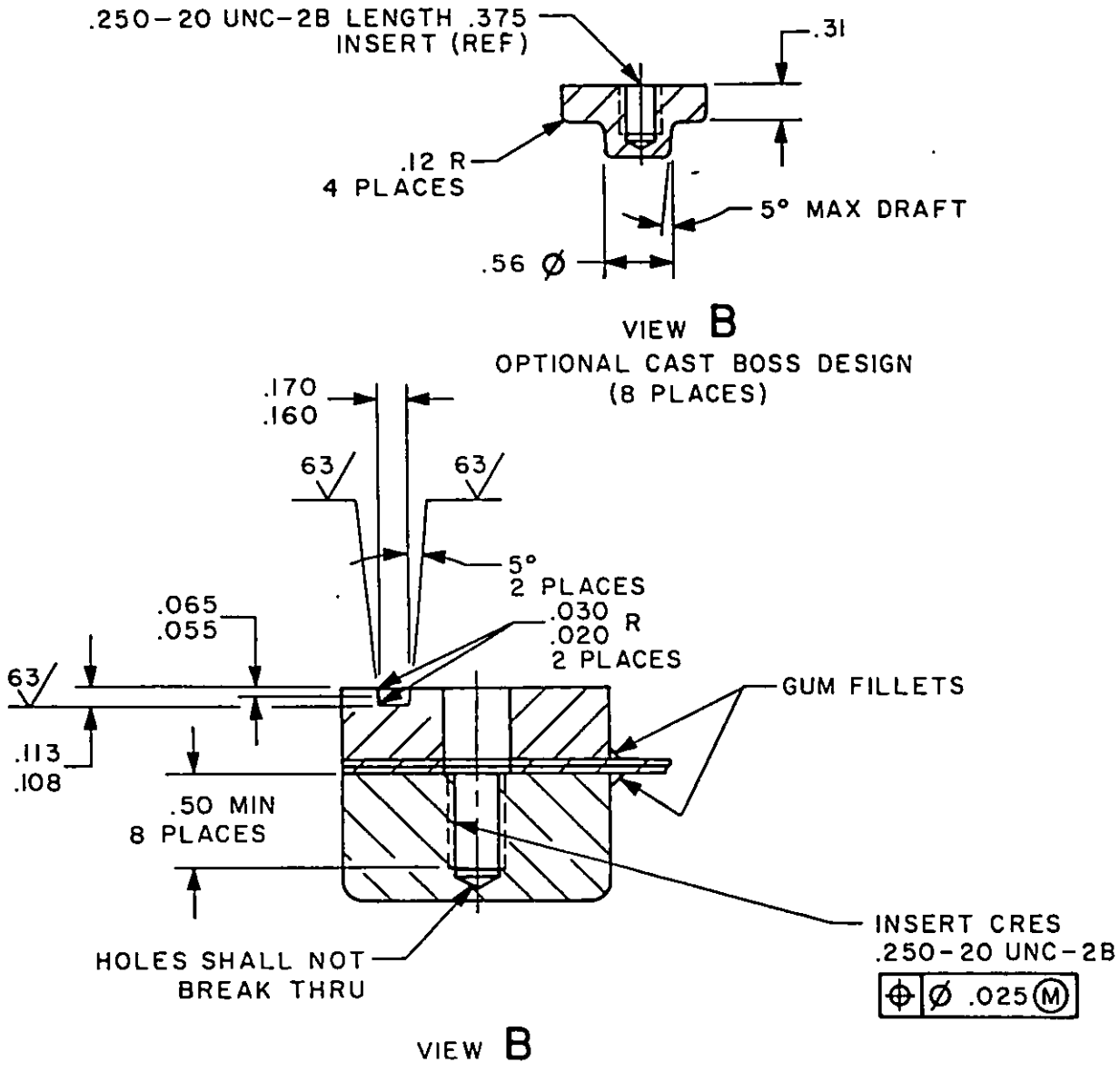


FIGURE 6a. Vent and drain fitting-compression type.

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NOTES:

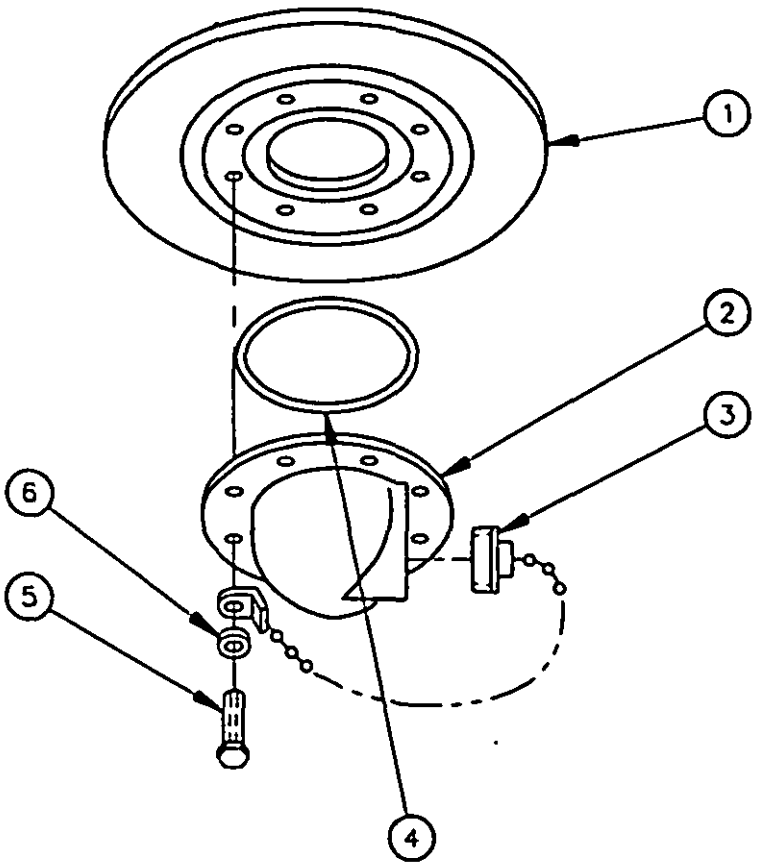
1. DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX = ± .06
   .XXX = ± .015
   ANGLES = ± 2°

2. ALUMINUM PARTS SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II, CLASS 2.

3. ADHESIVE BOND FABRIC AND RINGS.

FIGURE 6b. Vent and drain fitting-compression type views.

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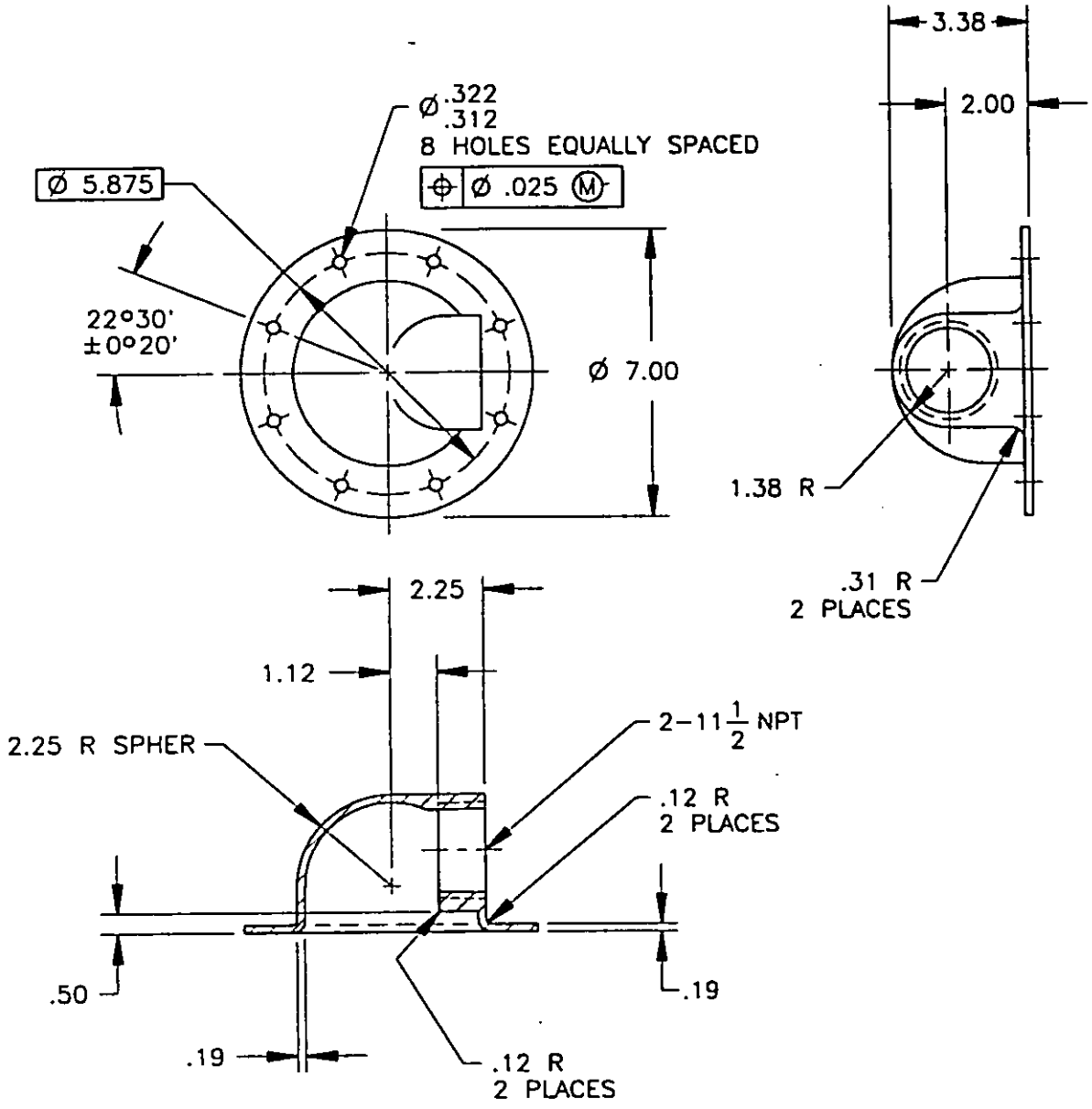
FIND NO.	PART NO. OR IDENT NO.	QTY	DESCRIPTION	SPECIFICATION
1	FIGURE 6	1	VENT AND DRAIN FITTING-COMPRESSION TYPE	
2	FIGURE 8	1	DRAIN FITTING	
3		1	PIPE PLUG, 2-INCH, CLASS 125	ANSI B16.14
4	MS29513-250	1	O-RING	MS29513
5		8	BOLT, HEX HD, 1/4-20 UNC-2A X 1.00 INCH LONG, STL, ZINC COATED, .003 IN. THK	ANSI B18.2.1
6		8	WASHER, PLAIN, .281 ID STL, ZINC COATED, .003 IN. THK	ANSI B18.22.1

NOTES:

1. LUBRICATE O-RING, FIND NO. 4, WITH SILICONE COMPOUND BEFORE ASSEMBLING HARDWARE.

FIGURE 7. Drain fitting assembly.

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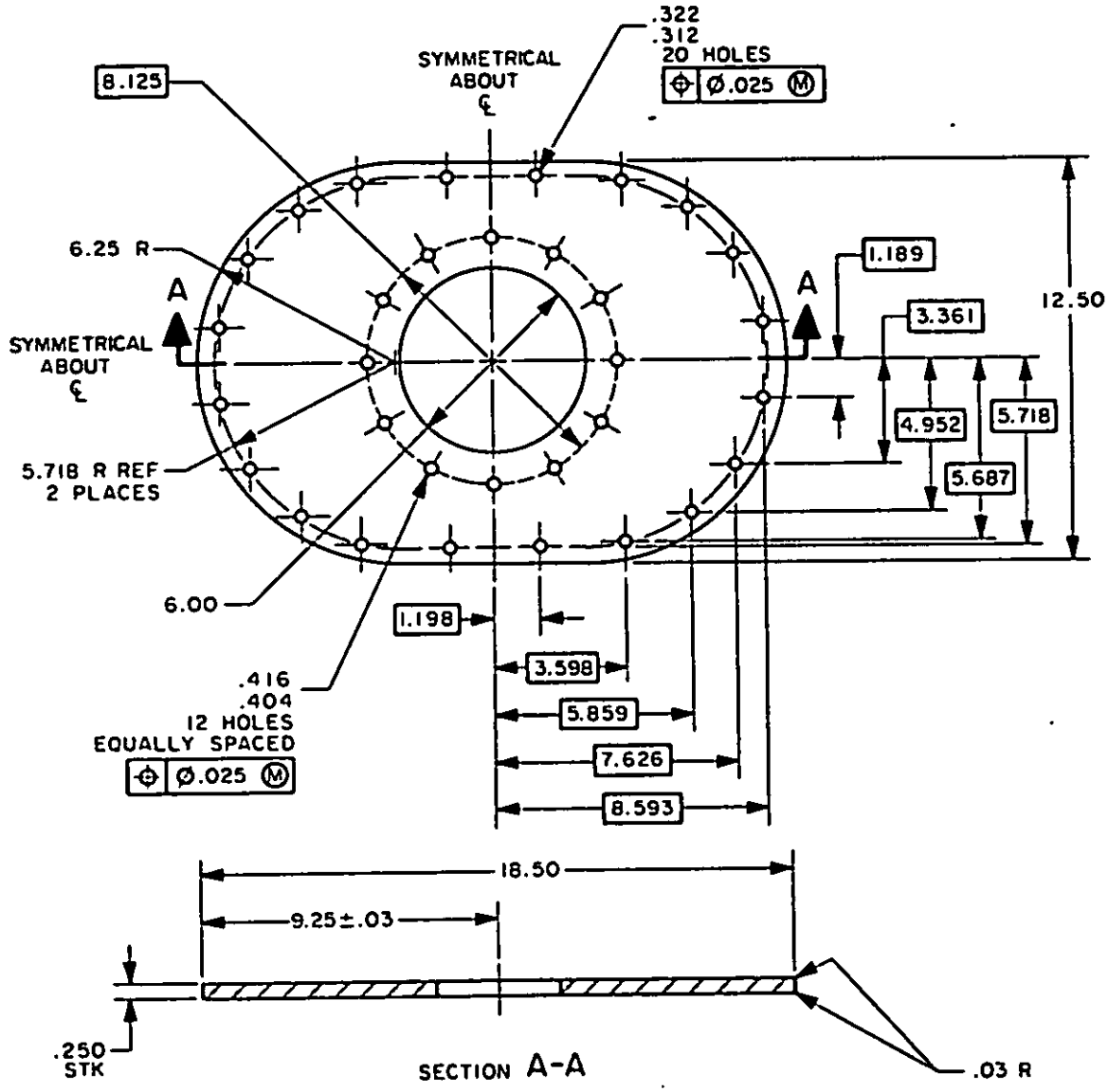


NOTES:

1. MATERIAL: AL ALLOY IN ACCORDANCE WITH ASTM B 26, UNS A13560-T6.
2. DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX ± .06
   DRAFT ANGLES = 3° MAX
3. ALUMINUM ALLOY PART SHALL BE ANODIZED IN ACCORDANCE WITH MIL-A-8625, TYPE II.

FIGURE 8. Drain fitting.

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NOTES:

1. DIMENSIONS ARE IN INCHES.
   TOLERANCES: .XX ± .06
   .XXX ± .015
   ANGLES ± 2° 0"
2. MATERIAL: AL ALLOY.
3. ANODIZE AND DYE IN ACCORDANCE WITH MIL-A-8625, TYPE II, CLASS 2.

FIGURE 9. Oval closure plate-compression fitting.

X-4128B

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NOTES:

1. MATERIAL: RUBBER, ASTM D2000 M4AG610A13E04.
2. DIMENSIONS ARE IN INCHES. TOLERANCES: ±.03.

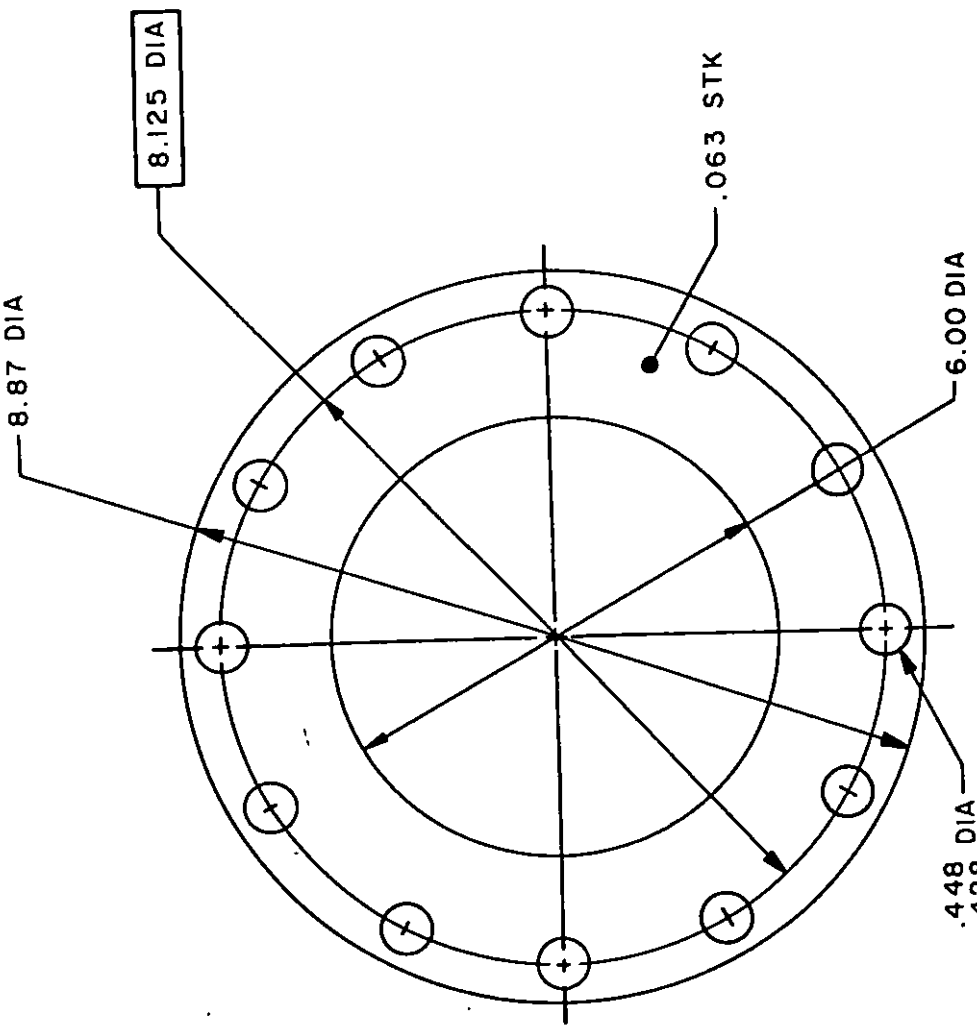


FIGURE 10. Gasket, for 6-inch tank truck round flange.

X-4127B

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STEP 1 REMOVE THE DRAIN HOSES AND ATTACH THE DUST PLUGS TO THE DRAIN FITTINGS. REMOVE THE FILLER/DISCHARGE ASSEMBLIES AND THE VENT ASSEMBLY. UTILIZING THE DEPLOYMENT STRAPS AND THE TANK HANDLES, MAKE THE TANK AS FLAT AS POSSIBLE. THE TANK SHOULD BE LAYING SMOOTH WITHOUT FOLDS AND WRINKLES BEFORE THE FOLDING PROCESS BEGINS (SEE FIGURE 12 STEP 1).

STEP 2 FOLD SIDE A (SEE FIGURE 13 STEP 2) TOWARD THE CENTER OF THE TANK BRINGING THE FOLDED SIDE FLUSH WITH THE VENT OPENING. MAKE SURE ACCESS TO THE VENT OPENING IS ALWAYS MAINTAINED. CONTINUE TO REMOVE AIR FROM THE TANK BY EVACUATING WITH A PUMP AND WALKING BACK AND FORTH ACROSS THE TANK AFTER IT IS FOLDED.

STEP 3 FOLD SIDE A (SEE FIGURE 13 STEP 3) AGAIN TOWARD THE CENTER OF THE TANK BRINGING THE SECOND FOLD FLUSH WITH THE VENT OPENING. MAKE SURE THAT ACCESS TO THE VENT OPENING IS ALWAYS MAINTAINED. CONTINUE TO REMOVE AIR FROM THE TANK BY EVACUATING WITH A PUMP AND WALKING BACK AND FORTH ACROSS THE TANK AFTER IT IS FOLDED.

STEP 4 REPEAT STEP 2 AND 3 FOR SIDE B.

STEP 5 AFTER SIDE A AND B ARE PROPERLY FOLDED BEGIN ROLLING FROM SIDE C (SEE FIGURE 14 STEP 5). ENSURE THAT THE VENT OPENING IS CLEAR AS IT MAY BE NECESSARY TO EVACUATE AIR FROM THE TANK AS YOU ROLL THE TANK. ATTACH DUST CAP TO VENT OPENING. ROLL AT LEAST ONE FULL ROLL PAST THE VENT OPENING.

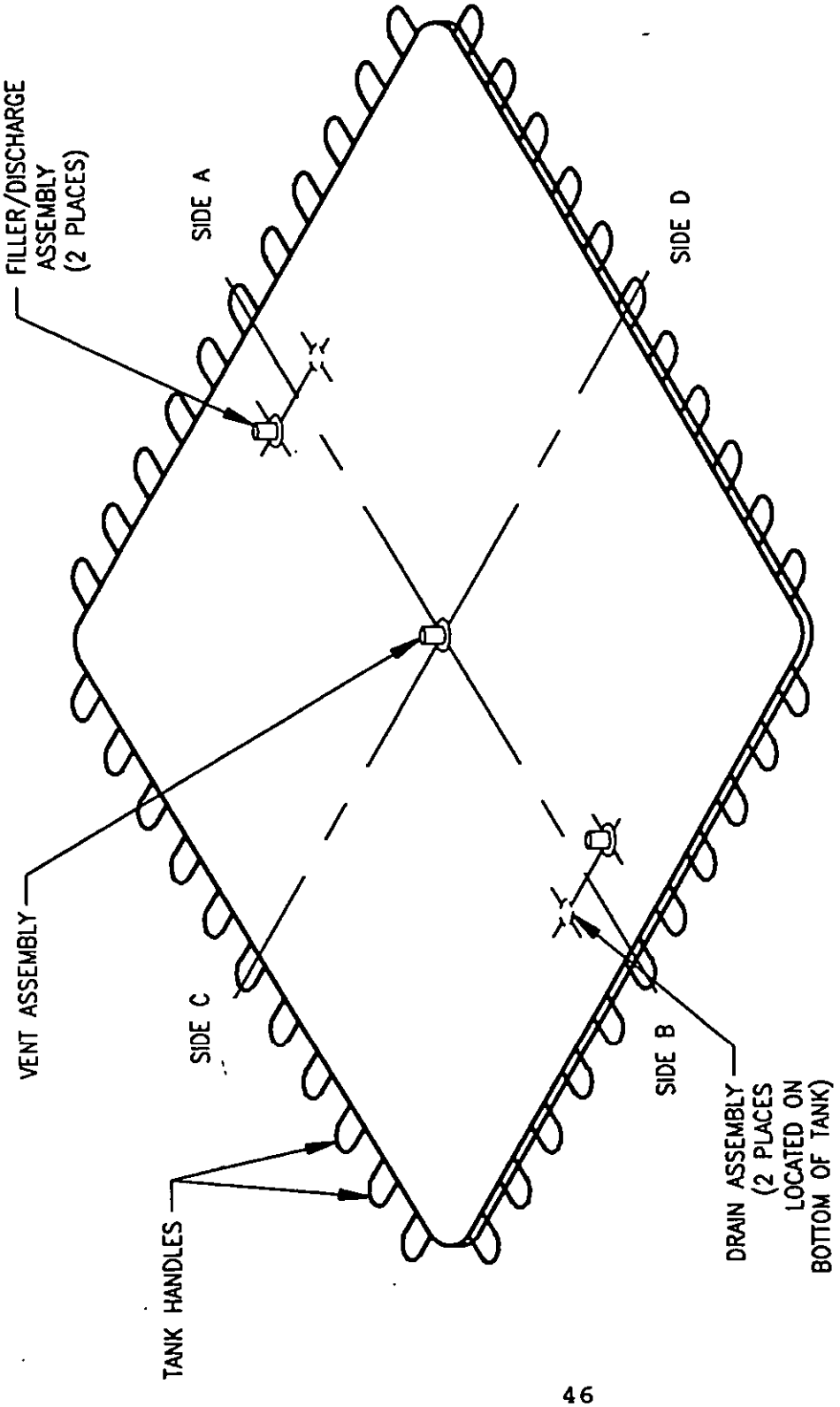
STEP 6 PLACE THE TANK SKID UP AGAINST THE ROLLED UP SIDE C (SEE FIGURE 14 STEP 6). PLACE LIFTING STRAPS ACROSS THE PALLET SO THEY MAY BE UTILIZED WHEN YOU SECURE THE TANK.

STEP 7 ROLL THE FOLDED END OF TANK BACKWARD ON THE PALLET UNTIL THE VENT OPENING IS CLEAR (SEE FIGURE 15 STEP 7). THEN BEGIN ROLLING UP SIDE D FOLLOWING THE PROCEDURES IN STEP 5.

STEP 8 WHEN THE FOLDING AND ROLLING OF THE TANK IS COMPLETE THE TANK SHOULD FIT WITHIN THE DIMENSIONS OF THE SKID. TO PREVENT THE TANK FROM UNROLLING UTILIZE THE LIFTING STRAPS TO SECURE THE TANK (SEE FIGURE 15 STEP 8).

FIGURE 11. Steps to roll tank.

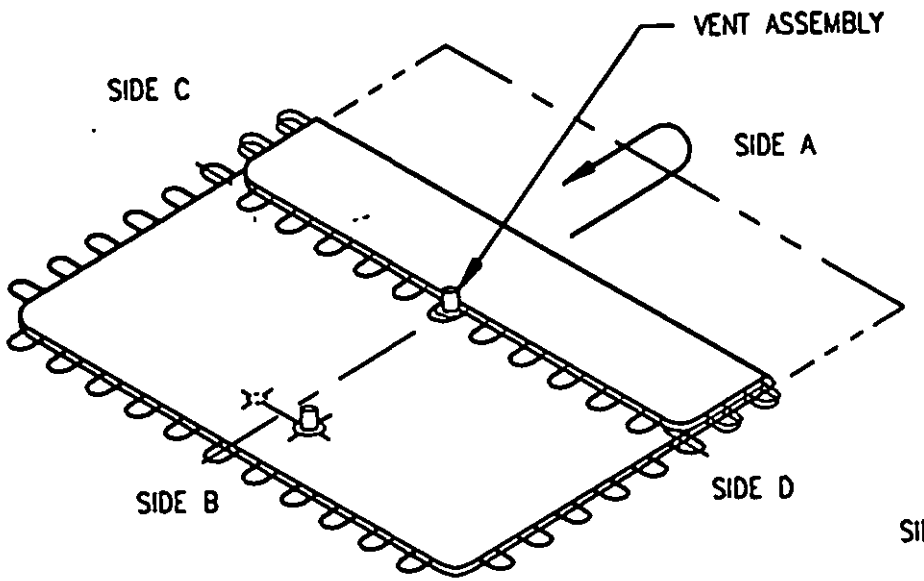
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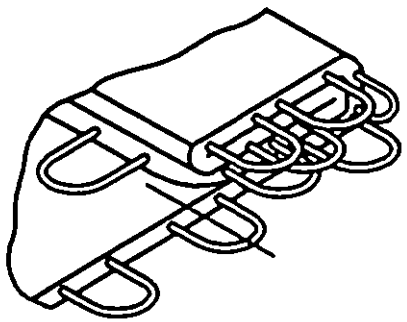
X-5043

FIGURE 12. 5,000 Barrel tank folding procedure.

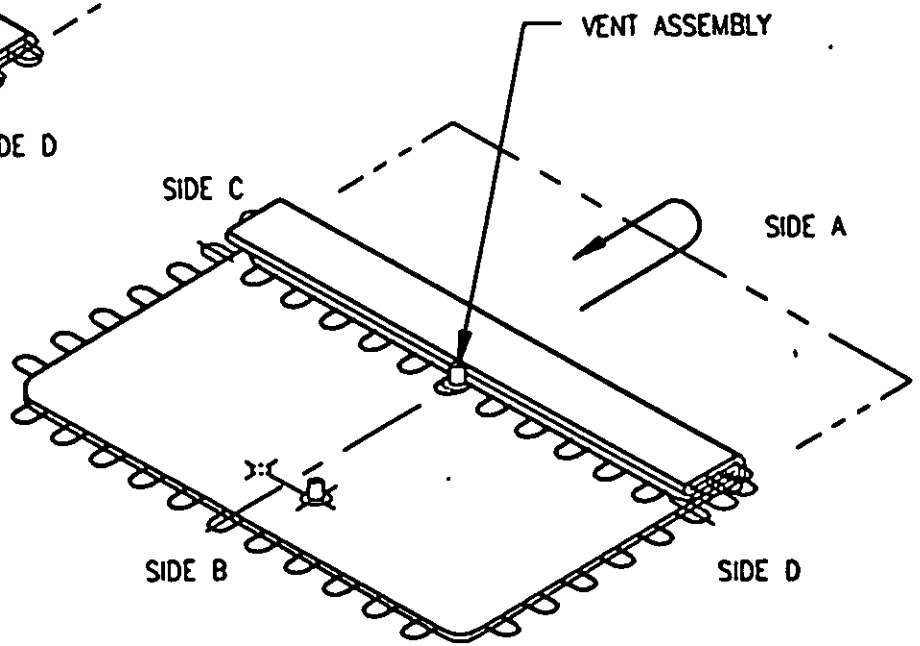
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STEP 2



STEP 3-SIDE A
ENLARGED VIEW - DOUBLE FOLDED



STEP 3

FIGURE 13. 5,000 Barrel tank folding procedure.

X-5044

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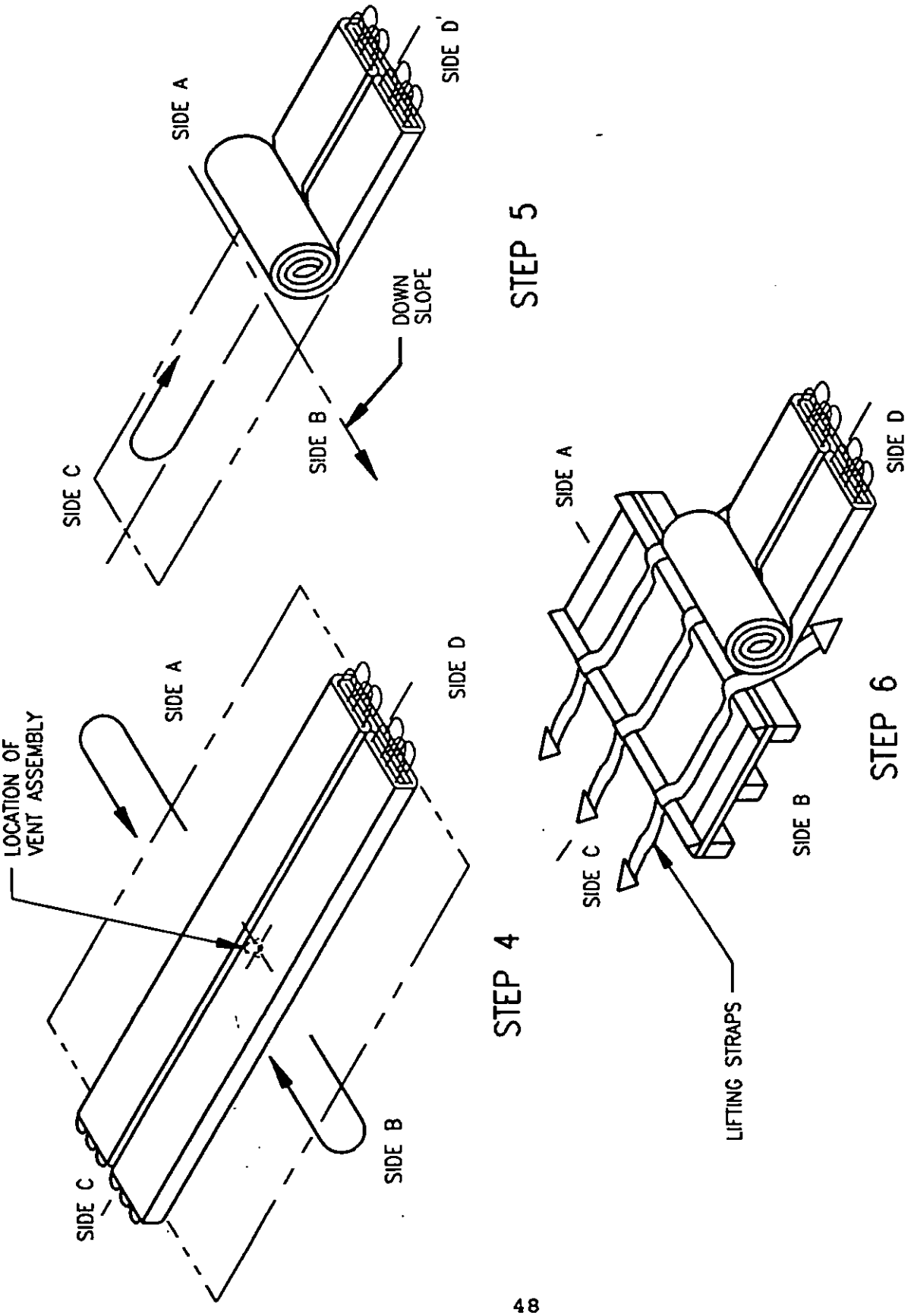
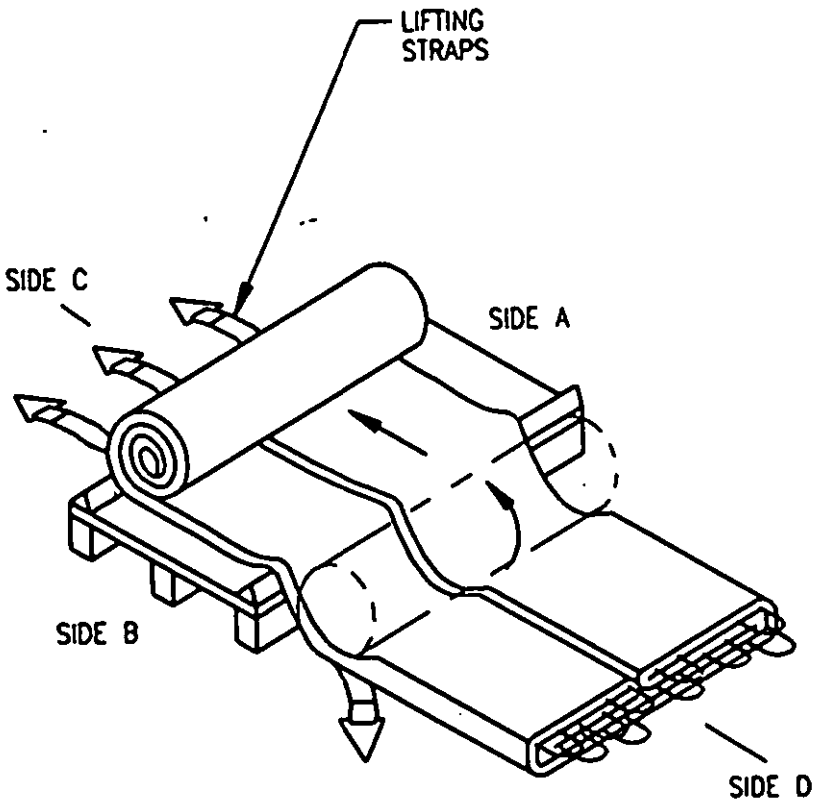


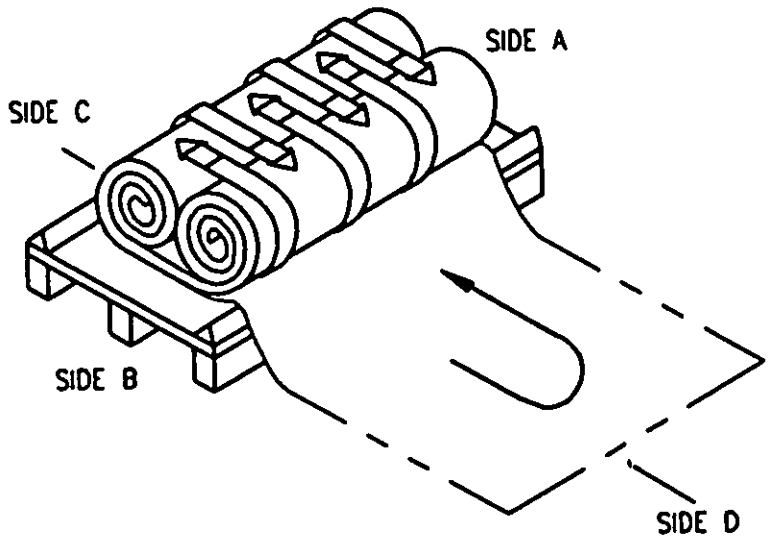
FIGURE 14. 5,000 Barrel tank folding procedure.

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STEP 7

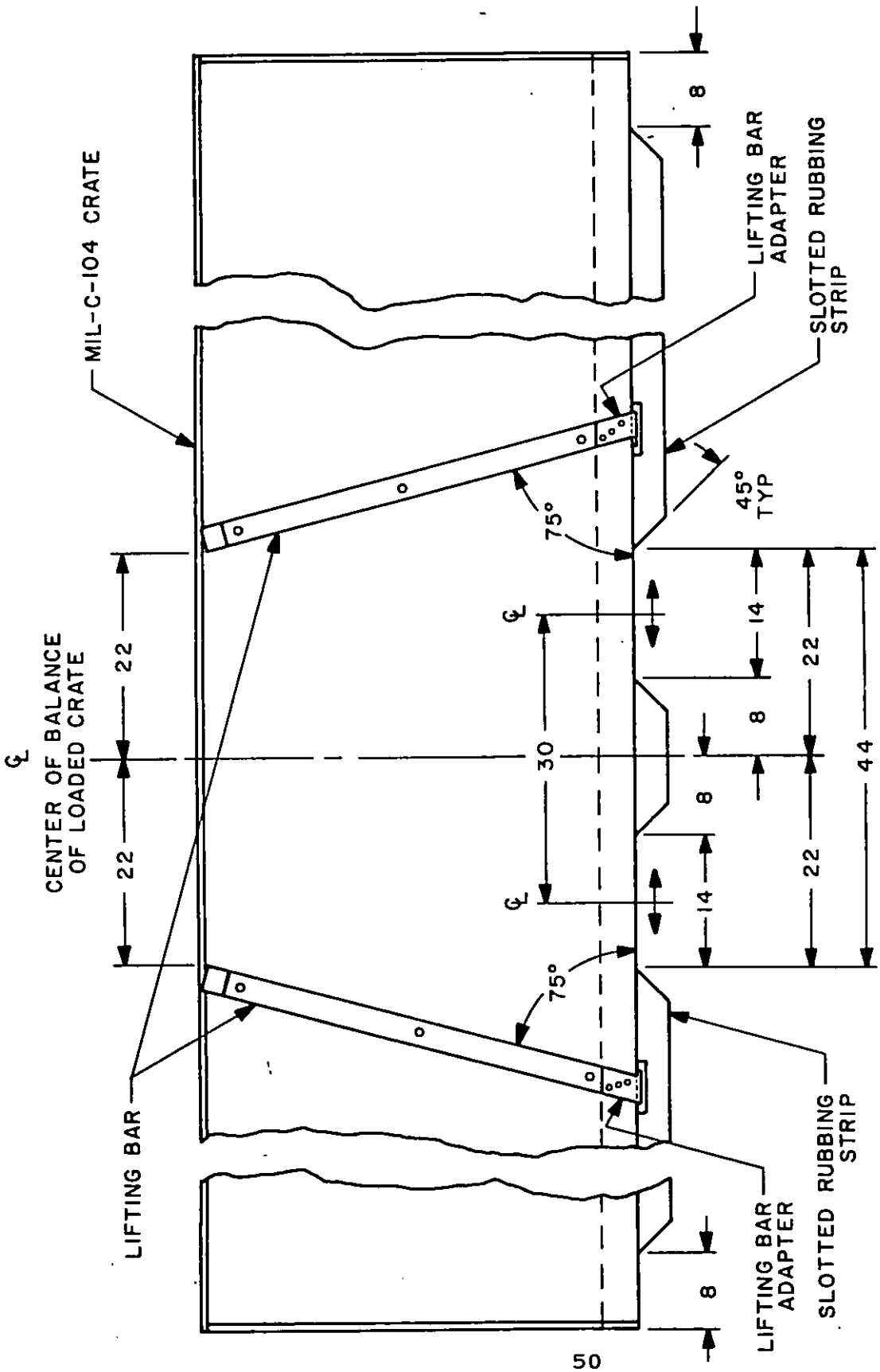


STEP 8

FIGURE 15. 5,000 Barrel tank folding procedure.

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X-4685

FIGURE 16. Lifting device positioning (Representative view of crate)

NOTE:
DIMENSIONS IN INCHES

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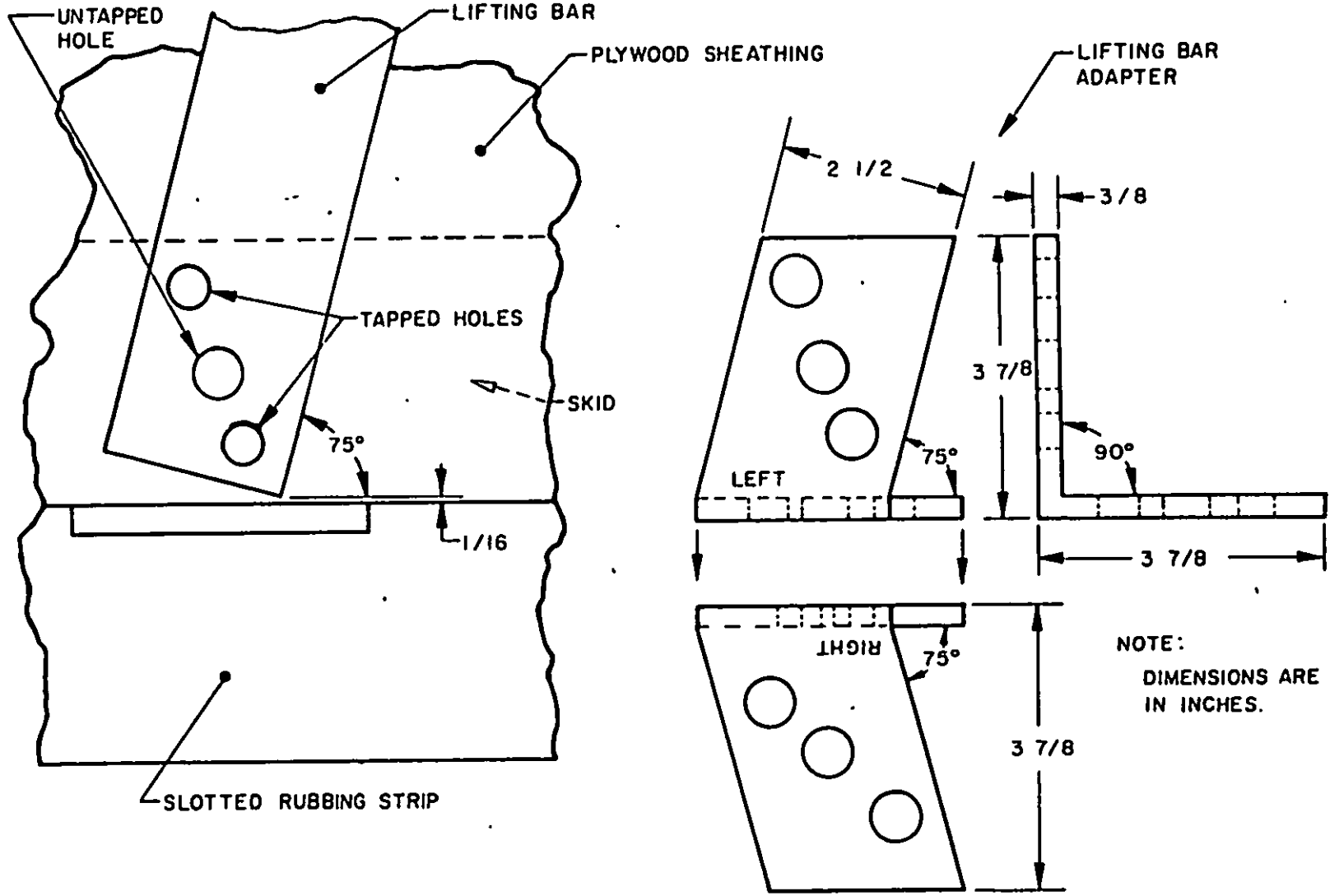
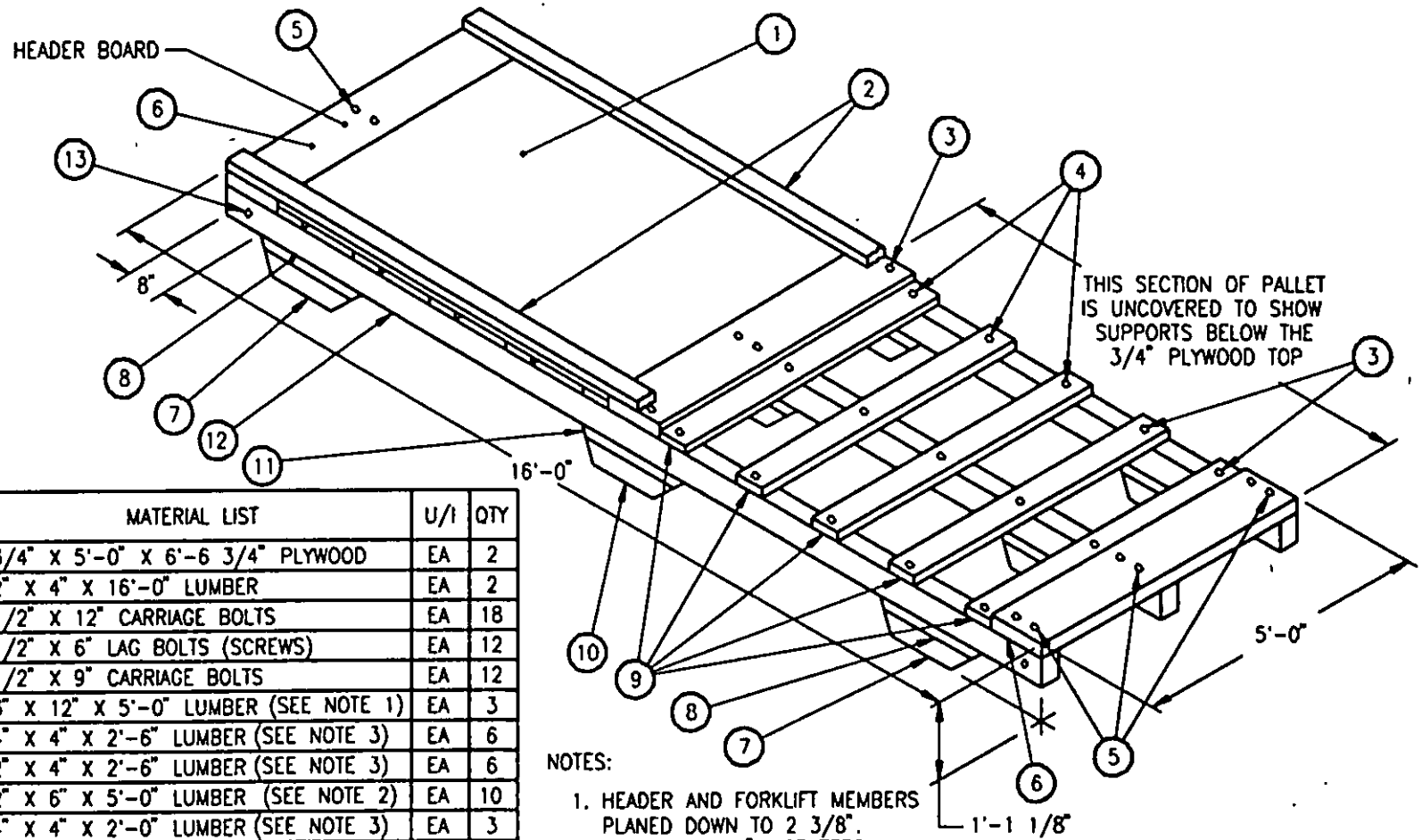


FIGURE 17. Lifting bar and Lifting bar adapter details.

X-4686

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FIND NO.	MATERIAL LIST	U/I	QTY
1	3/4" X 5'-0" X 6'-6 3/4" PLYWOOD	EA	2
2	2" X 4" X 16'-0" LUMBER	EA	2
3	1/2" X 12" CARRIAGE BOLTS	EA	18
4	1/2" X 6" LAG BOLTS (SCREWS)	EA	12
5	1/2" X 9" CARRIAGE BOLTS	EA	12
6	3" X 12" X 5'-0" LUMBER (SEE NOTE 1)	EA	3
7	4" X 4" X 2'-6" LUMBER (SEE NOTE 3)	EA	6
8	2" X 4" X 2'-6" LUMBER (SEE NOTE 3)	EA	6
9	2" X 6" X 5'-0" LUMBER (SEE NOTE 2)	EA	10
10	4" X 4" X 2'-0" LUMBER (SEE NOTE 3)	EA	3
11	2" X 4" X 2'-0" LUMBER (SEE NOTE 3)	EA	3
12	4" X 6" X 16'-0" LUMBER	EA	3
13	1/2" X 5" CARRIAGE BOLTS	EA	6
14	NAILS 16d (NOT SHOWN ON DRAWING)	LB	1

NOTES:

1. HEADER AND FORKLIFT MEMBERS PLANED DOWN TO 2 3/8".
2. PLACED ON 16" + CENTERS.
3. BEVEL 45° AS SHOWN.
4. COUNTER SINK ALL CARRIAGE BOLTS.
5. MATERIAL LIST DIMENSIONS ARE ALL NOMINAL, PLAN SKID DIMENSIONS ARE ACTUAL.

FIGURE 18. Skid for 5,000 bbl. tank.

X-5047

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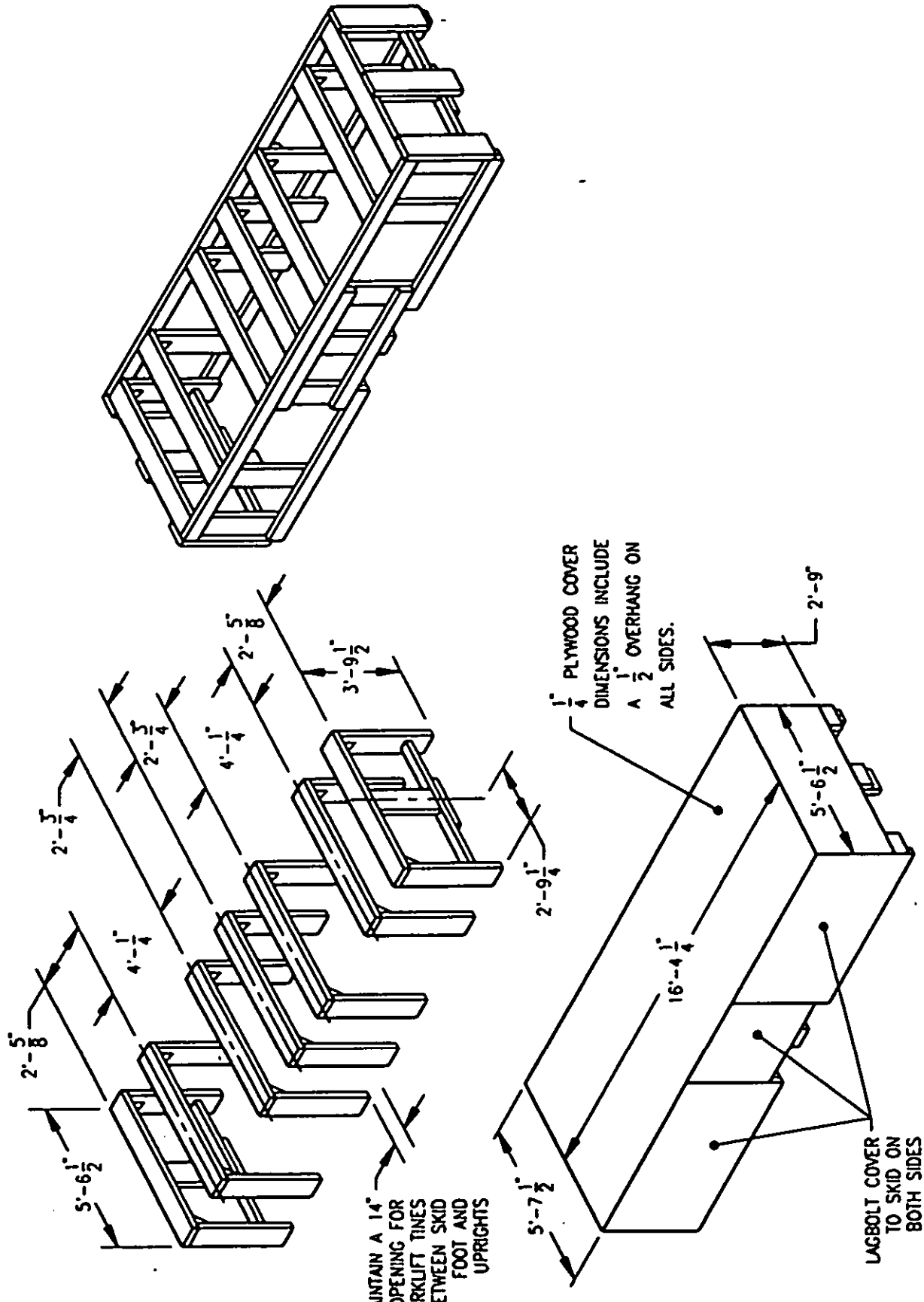
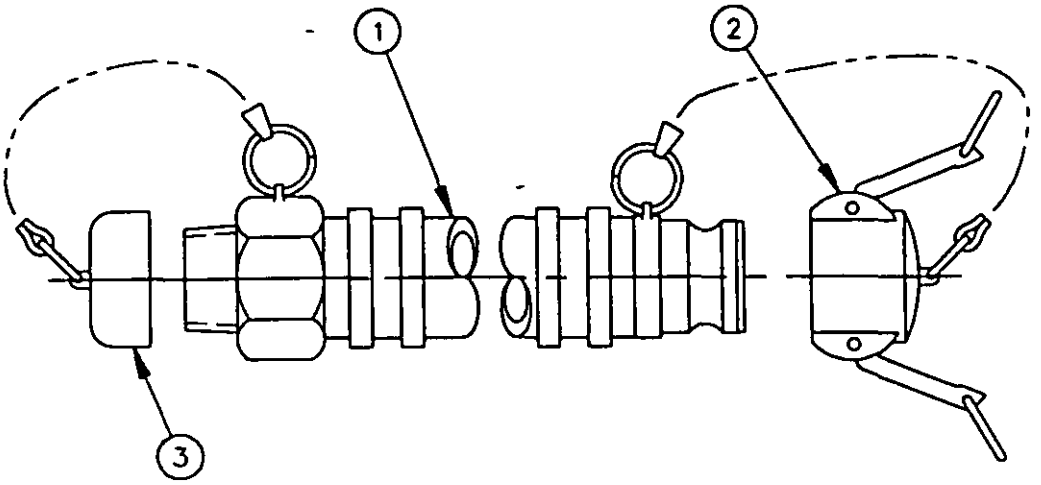


FIGURE 19. 5,000 Barrel tank skid cover.

X-5048

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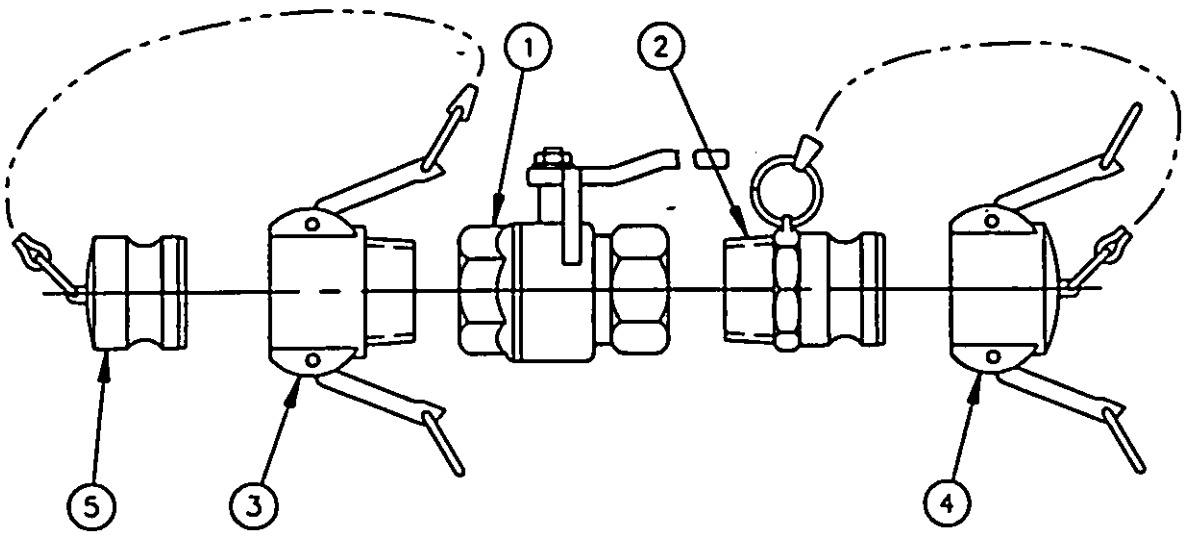
FIND NO.	PART NO. OR IDENT NO.	QTY REQD		DESCRIPTION	SPECIFICATION
		FUEL TANK	WATER TANK		
1	M370B06C21200	1	–	HOSE ASSEMBLY, TYPE II, SIZE 6 (2-INCH), CLASS 2, STYLE A, 10-FOOT LENGTH, WITH 2-INCH ALUMINUM MALE (NPT) X CAM-LOCKING MALE BANDED SHANK FITTINGS ON BOTH ENDS	MIL-H-370
1	ZZH561BA2020120	–	1	HOSE ASSEMBLY, GRADE B, CLASS 2, SIZE 2-INCH, 10-FOOT LENGTH, WITH 2-INCH BRASS MALE (NPT) X CAM-LOCKING MALE BANDED SHANK FITTINGS ON BOTH ENDS	ZZ-H-561
2	MS27028-11	1	–	DUST CAP	–
2	MS27028-12	–	1	DUST CAP	–
3		1	1	THREADED PIPE CAP, 2-INCH	ANSI B16.15

NOTES:

1. LUBRICATE GASKET, PART OF FIND NO. 2, WITH SILICONE COMPOUND PRIOR TO ASSEMBLY.

FIGURE 20. Hose assembly (Drain).

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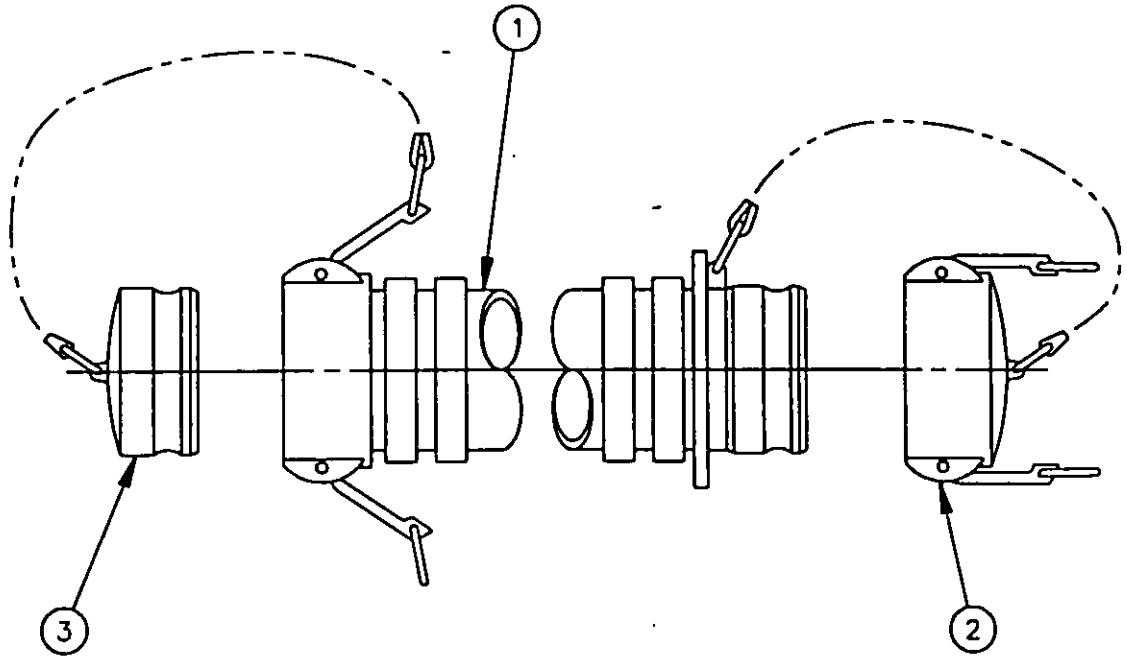
FIND NO.	PART NO. OR IDENT NO.	QTY REQD		DESCRIPTION	SPECIFICATION
		FUEL TANK	WATER TANK		
1		1	–	BALL VALVE, TYPE II, COMPOSITION CS, STYLE 1, END CONNECTION THREADED, CLASS 125, SIZE 2	WW-V-35
1		–	1	BALL VALVE, TYPE II, COMPOSITION BZ, STYLE 1, END CONNECTION THREADED, CLASS 125, SIZE 2	WW-V-35
2	MS27022-11	1	–	COUPLING HALF	
2	MS27022-12	–	1	COUPLING HALF	
3	MS27026-11	1	–	COUPLING HALF	
3	MS27026-12	–	1	COUPLING HALF	
4	MS27028-11	1	–	DUST CAP	
4	MS27028-12	–	1	DUST CAP	
5	MS27029-11	1	–	DUST PLUG	
5	MS27029-12	–	1	DUST PLUG	

NOTES:

1. LUBRICATE GASKET, PART OF FIND NO. 3 AND 4, WITH SILICONE COMPOUND PRIOR TO ASSEMBLY.

FIGURE 21. Ball valve assembly (Drain).

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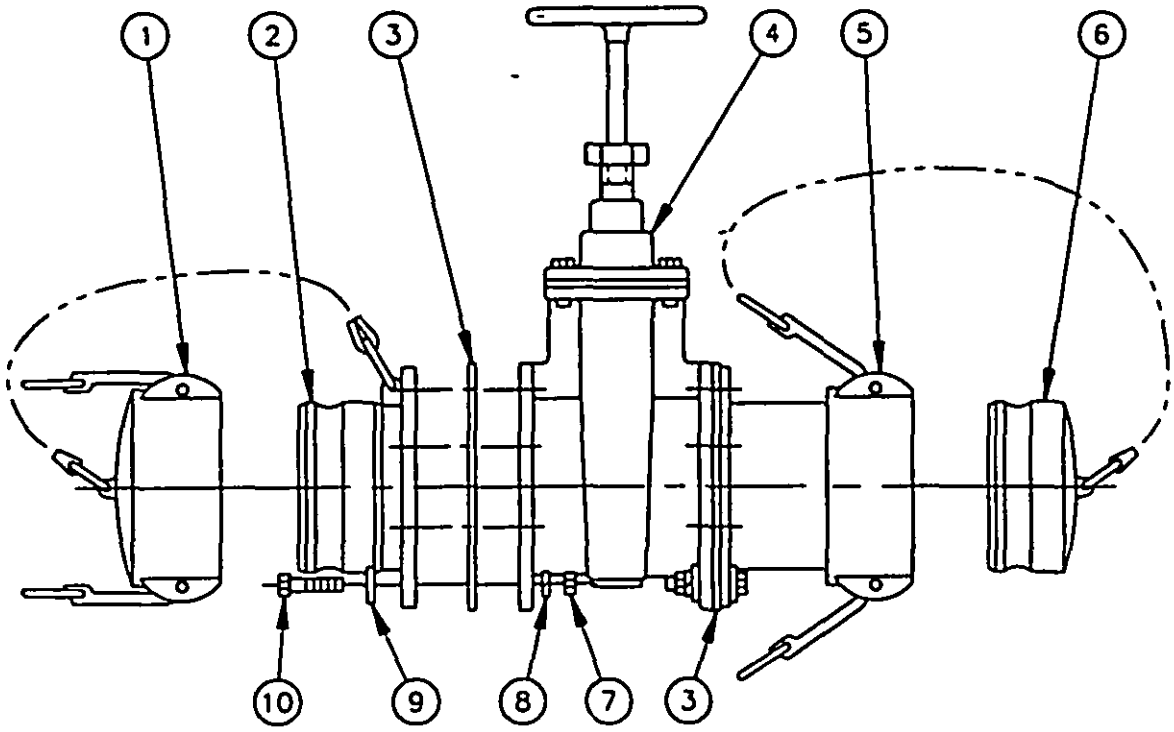
FIND NO.	PART NO. OR IDENT NO.	QTY REQD	DESCRIPTION	SPECIFICATION
1	M370B10B21200	1	HOSE ASSEMBLY, TYPE II, CLASS 1, SIZE 10 (6-INCH), STYLE A, 10-FOOT LENGTH WITH 6-INCH ALUMINUM CAM LOCKING MALE X FEMALE BANDED SHANK FITTINGS	MIL-H-370
2	MS27028-19	1	DUST CAP	
3	MS27029-19	1	DUST PLUG	

NOTES:

1. LUBRICATE GASKET, PART OF FIND NO. 1 AND 2, WITH SILICONE COMPOUND PRIOR TO ASSEMBLY.

FIGURE 22. Hose assembly (Filler/Discharge).

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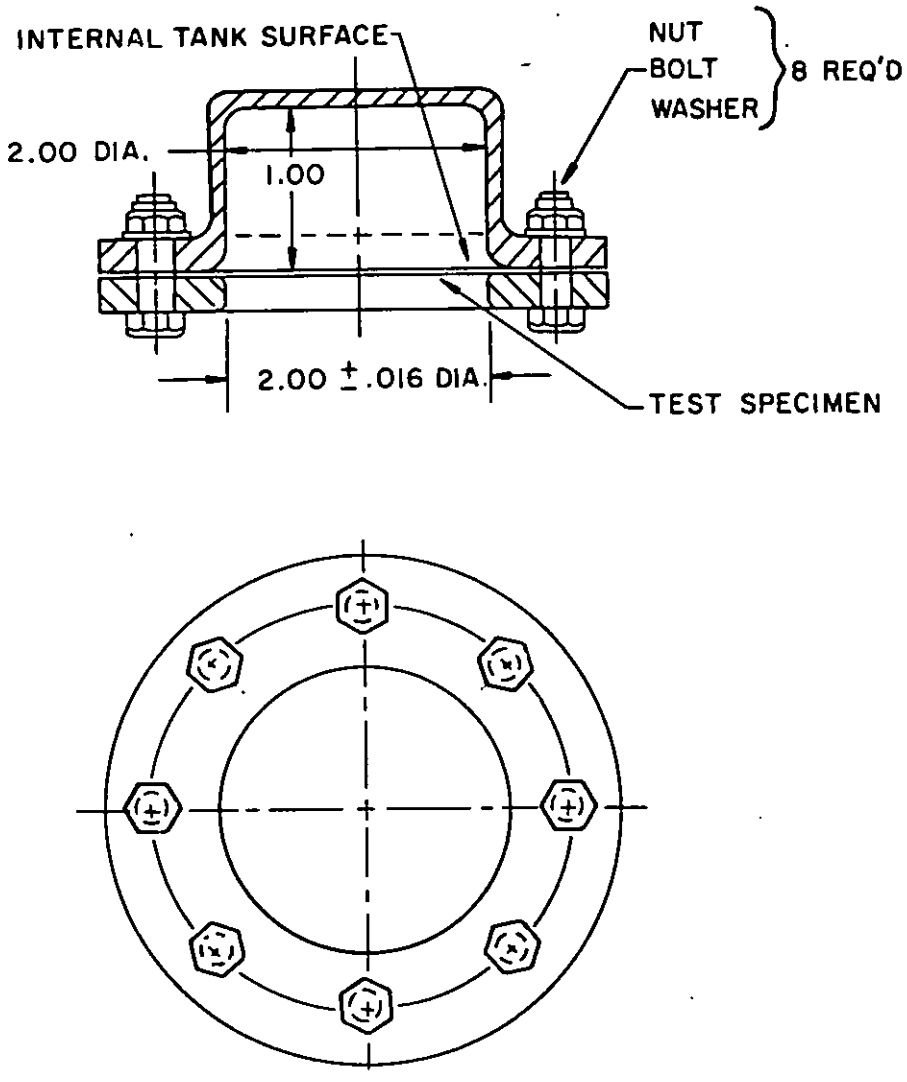
FIND NO.	PART NO. OR IDENT NO.	QTY	DESCRIPTION	SPECIFICATION
1	MS27028-19	1	DUST CAP	
2	MS27023-19	1	COUPLING HALF	
3	FIGURE 10	2	GASKET	
4		1	GATE VALVE, SIZE 6-INCH, TYPE I	MIL-V-58039
5	MS27027-19	1	COUPLING HALF	
6	MS27029-19	1	DUST PLUG	
7		24	MACHINE SCREW HEX NUT, .375-INCH	ANSI B18.2.2
8		24	HELICAL LOCK WASHER, .375-INCH	ANSI B18.21.1
9		24	PLAIN WASHER, .375-INCH	ANSI B18.22.1
10		24	HEX BOLT, .375-INCH	ANSI B18.2.1

NOTES:

1. LUBRICATE GASKETS, FIND NO. 3 AND PART OF FIND NO. 1 AND 5, WITH SILICONE COMPOUND BEFORE ASSEMBLING HARDWARE.

FIGURE 23. Gate valve assembly (Filler/Discharge).

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NOTES:

1. DIMENSIONS ARE IN INCHES.

   TOLERANCES: .XX ± .06

FIGURE 24. Diffusion cup assembly.

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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-530668	2. DOCUMENT DATE (YYMMDD) 940928

3. DOCUMENT TITLE
TANK, FABRIC, COLLAPSIBLE, 5,000 BARREL, FUEL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468	(2) DSN 654-3468
c. ADDRESS (Include Zip Code) US ARMY BELVOIR RDE CTR ATTN AMSTA-RBE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340	DSN 289-2340

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